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Mas-Colell Solutions

Chapter 1

Question 1.B.1. Prove that if $\succeq$ is rational then if $x \succ y \succeq z$, then $x \succ z$.

Answer 1.B.1. (Proof by contradiction). Suppose $x \succ y \succeq z$ and $z \succeq x$ from some $x, y, z \in X$. Then $y \succeq z \succeq x$; by transitivity, $x \succeq y$, contradicting our assumption that $x \succ y$. Therefore, $x \succ z$.

Question 1.B.2. Prove that if $\succeq$ is rational then

- (1) $\succ$ is both irreflexive ($x \succ x$ never holds) and transitive (if $x \succ y$ and $y \succ z$, then $x \succ z$)
- (2) $\sim$ is reflexive ($x \sim x$ for all x), transitive (if $x \sim y$ and $y \sim z$, then $x \sim z$), and symmetric (if $x \sim y$, then $y \sim x$).

Answer 1.B.2. Proof of (1):

- (Irreflexivity) Suppose $\exists x \in X$ such that $x \succ x$. Then by definition of $\succ$, we have $x \succeq x$ but not $x \succeq x$, which is obviously silly.
- (Transitivity) Suppose $x \succ y$ and $y \succ z$. Then $x \succeq y$, but not $y \succeq x$; and $y \succeq z$, but not $z \succeq y$. Then by transitivity of $\succeq$, $x \succeq z$. However, $z \succeq x$ is impossible, since if it were, then together with $y \succeq z$, transitivity would imply $y \succeq x$, contradicting $x \succ y$. Therefore, $x \succeq z$ but not $z \succeq x$, i.e. $x \succ z$.

Proof of (2):

- (Reflexivity) For all $x \in X$, $x \succeq x$ and $x \succeq x$, i.e. $x \sim x$.
- (Transitivity) Suppose $x, y, z \in X$, $x \sim y$, and $y \sim z$, i.e., $x \succeq y$, $y \succeq x$, $y \succeq z$, and $z \succeq y$. Then by transitivity, $x \succeq z$ and $z \succeq x$, i.e. $x \sim z$.
- (Symmetry) Suppose $x, y \in X$ and $x \sim y$. Then $x \succeq y$ and $y \succeq x$, i.e. $y \sim x$.

Question 1.B.3. Show that if $f : \mathbb{R} \rightarrow \mathbb{R}$ is a strictly increasing function and $u : X \rightarrow \mathbb{R}$ is a utility function representing preference relation $\succeq$, then the function $v : X \rightarrow \mathbb{R}$ defined by $v(x) = f(u(x))$ is also a utility function representing preference relation $\succeq$.

Answer 1.B.3. Suppose $x, y \in X$ and $x \succeq y$. Since u represents $\succeq$, $u(x) \geq u(y)$. Since f is strictly increasing, $f(u(x)) \geq f(u(y))$, i.e. $v = f \circ u$ represents $\succeq$.

Question 1.B.4. Consider a rational preference relation $\succeq$. Show that if $u(x) = u(y)$ implies $x \sim y$ and if $u(x) > u(y)$ implies $x \succ y$, then $u(\cdot)$ is a utility function representing $\succeq$.

Answer 1.B.4. Suppose $x, y \in X$, $x \succeq y$, but $u(x) < u(y)$. Then $x \succeq y$, and $y \succ x$, which is impossible. So, $x \succeq y$ implies $u(x) \geq u(y)$, i.e., u represents $\succeq$.

Question 1.B.5. Show that if X is finite and $\succeq$ is a rational preference relation on X , then there is a utility function $u : X \rightarrow \mathbb{R}$ that represents $\succeq$. [*Hint:* Consider first the case in which the individual's ranking between any two elements of X is strict (i.e., there is never any indifference), and construct a utility function representing those preferences; then extend your argument to the general case.]

Answer 1.B.5. Note: $|S|$ represents the number of elements of a finite set S .

Since X is finite, $|X| = n$ for some $n \in \mathbb{Z}_{>0}$. Now, define $u : X \rightarrow \{1, 2, \dots, n\}$ by $x \rightarrow |\{\bar{x} \in X | x \succeq \bar{x}\}|$. We claim that u represents $\succeq$. Suppose $x \succeq y$. Then $u(x) = |\{\bar{x} \in X | x \succeq \bar{x}\}|$ and $u(y) = |\{\bar{x} \in X | y \succeq \bar{x}\}|$. Since $x \succeq y$, $x_0 \in \{\bar{x} \in X | y \succeq \bar{x}\}$ implies $x_0 \in \{\bar{x} \in X | x \succeq \bar{x}\}$ but the implication fails in the other direction. Since both sets are finite, $u(x) \geq u(y)$, i.e. u represents $\succeq$.

Question 1.C.1. Consider the choice structure $(\mathcal{B}, C(\cdot))$ with $\mathcal{B} = (\{x, y\}, \{x, y, z\})$ and $C(\{x, y\}) = \{x\}$. Show that if $(\mathcal{B}, C(\cdot))$ satisfies the weak axiom, then we must have $C(\{x, y, z\}) = \{x\}$, $= \{z\}$, or $= \{x, z\}$.

Answer 1.C.1. (Proof by contradiction) Suppose $y \in C(\{x, y, z\})$. Then since $x, y \in \{x, y\}$ implies $x, y \in C(\{x, y\})$ by the weak axiom, a contradiction. Therefore, $C(\{x, y, z\})$ must be a subset of $\{x, y, z\}$ not containing y , namely one of $\{x\}$, $\{z\}$, and $\{x, z\}$.

Question 1.C.2. Show that the weak axiom (Definition 1.C.1) is equivalent to the following property holding:

Suppose that $B, B' \in \mathcal{B}$, that $x, y \in B$ and that $x, y \in B'$. Then in $x \in C(B)$ and $y \in C(B')$, we must have $\{x, y\} \subset C(B)$ and $\{x, y\} \subset C(B')$.

Answer 1.C.2. The forward direction (the property in the problem implies the weak axiom) is clear, since the property in the problem is just a restatement of the weak axiom. The backward direction (the weak axiom implies the property in the problem) follows from applying the weak axiom first to B' and then to B to show $y \in C(B)$, $x \in C(B')$.

Question 1.C.3. Suppose that choice structure $(\mathcal{B}, C(\cdot))$ satisfies the weak axiom. Consider the following two possible reveal preferred relations, $\succ^*$ and $\succ^{**}$:

$$\begin{aligned} x \succ^* y &\Leftrightarrow \text{there is some } B \in \mathcal{B} \text{ such that } x, y \in B, x \in C(B), \text{ and } y \notin C(B) \\ x \succ^{**} y &\Leftrightarrow x \succeq^* y \text{ but not } y \succeq^* x \end{aligned}$$

where $\succeq^*$ is the revealed at-least-as-good-as relation defined in Definition 1.C.2.

Question 1.C.3.a. Show that $\succ^*$ and $\succ^{**}$ give the same relation over X ; that is for any $x, y \in X$, $x \succ^* y \Leftrightarrow x \succ^{**} y$. Is this still true if $(\mathcal{B}, C(\cdot))$ does not satisfy the weak axiom?

Suppose x, y are any elements of X with $x \succ^* y$. Then there exists $B \in \mathcal{B}$ such that $x, y \in B$, $x \in C(B)$, but $y \notin C(B)$. In particular, this statement satisfies Definition 1.C.2, that $x \succeq^* y$ if there exists $B' \in \mathcal{B}$ where $x, y \in B'$ and $x \in C(B')$. However, clearly $y \not\succeq^* x$; if it were then there would exist $B' \in \mathcal{B}$ where $x, y \in B'$ and $y \in C(B')$. If so, then $x, y \in B$ with $x \in C(B)$ together with the weak axiom implies $y \in C(B)$, contradicting $x \succ^* y$. Therefore, $x \succeq^* y$ and $y \not\succeq^* x$, thus $x \succ^{**} y$.

Now suppose $x \succ^{**} y$, so that $x \succeq^* y$ but $y \not\succeq^* x$. In other words, there exists $B \in \mathcal{B}$ such that $x, y \in B$ with $x \in C(B)$. Also, for all $B' \in \mathcal{B}$ with $x, y \in B'$, $y \notin C(B')$. In particular, $y \notin C(B)$. So, there exists $B \in \mathcal{B}$ such that $x, y \in B$, $x \in C(B)$, and $y \notin C(B)$, thus $x \succ^* y$.

However, if the weak axiom does not hold, $\succ^*$ and $\succ^{**}$ may not be the same preference relation. Consider $\mathcal{B} = \{\{x, y\}, \{x, y, z\}\}$ with $C_2(\{x, y\}) = \{x\}$ and $C_2(\{x, y, z\}) = \{x, y\}$. (This choice structure is Mas-Colell's example of a choice structure that violates the weak axiom. See pages 10 - 11.) Since $x, y \in \{x, y\}$, $x \in C_2(\{x, y\})$, and $y \notin C_2(\{x, y\})$, $x \succ^* y$. However, $x \succeq^* y$ since $x, y \in \{x, y\}$ and $x \in C_2(\{x, y\})$ as well as $y \succeq^* x$ since $x, y \in \{x, y, z\}$ and $y \in C_2(\{x, y, z\})$, thus $x \not\succ^{**} y$.

Question 1.C.3.b. Must $\succ^*$ be transitive?

Answer 1.C.3.b. No. Consider the choice structure $\mathcal{B} = \{\{x, y\}, \{y, z\}\}$, $C(\{x, y\}) = x$, and $C(\{y, z\}) = y$. The choice structure (vacuously) satisfies the weak axiom because no two elements of $\mathcal{B}$ have two elements in common. Also, $x \succ^* y$ and $y \succ^* z$. However, it is impossible to compare x and z at all in this choice structure, since there is no $B \in \mathcal{B}$ containing both x and z . In particular, we cannot say that $x \succ^* z$.

Question 1.C.3.c. Show that if $\mathcal{B}$ includes all three-element subsets of X , then $\succ^*$ is transitive.

Answer 1.C.3.c. Suppose that $x \succ^* y$ and $y \succ^* z$. Then there is some $B_1 \in \mathcal{B}$ such that $x, y \in B_1$, $x \in C(B_1)$, $y \notin C(B_1)$, and there is some $B_2 \in \mathcal{B}$ such that $y, z \in B_2$, $y \in C(B_2)$, $z \notin C(B_2)$. Also, assuming the weak axiom, $x \succ^{**} y$ and $y \succ^{**} z$.

By assumption, $\{x, y, z\} \in \mathcal{B}$. We will show that $x \in C(\{x, y, z\})$ but $z \notin C(\{x, y, z\})$ which will complete the proof. First, suppose $z \in C(\{x, y, z\})$. Then $z \succeq^* y$, contradicting $y \succ^{**} z$. Thus, $z \notin C(\{x, y, z\})$.

Now, $C(\{x, y, z\})$ is non-empty, so $x \in C(\{x, y, z\})$ or $y \in C(\{x, y, z\})$. If $x \in C(\{x, y, z\})$ we are done. If $y \in C(\{x, y, z\})$, then $x \succ^{**} y$ implies $x \succeq^* y$, so by the weak axiom, $x \in C(\{x, y, z\})$.

Therefore, there exists $B = \{x, y, z\} \in \mathcal{B}$ such that $x \in C(B)$ but $z \notin C(B)$, so $x \succ^* z$, i.e. $\succ^*$ is transitive.

Question 1.D.1. Give an example of a choice structure that can be rationalized by several preference relations. Note that if the family of budgets $\mathcal{B}$ includes all the two-elements subsets of X , then there can be at most one rationalizing preference relation.

Answer 1.D.1. Put $X = \{x, y, z\}$, $\mathcal{B} = \{\{x, y, z\}, \{x, y\}\}$, and $C(\{x, y, z\}) = C(\{x, y\}) = \{x\}$. Let $\succeq^*$ be the relation $x \succ^* y$, $x \succ^* z$, $y \succ^* z$. Let $\succeq^{**}$ be the relation $x \succ^{**} y$, $x \succ^{**} z$, $z \succ^{**} y$. Then $C(\{x, y, z\}, \succ^*) = C(\{x, y\}, \succ^*) = C(\{x, y, z\}, \succ^{**}) = C(\{x, y\}, \succ^{**}) = \{x\}$ but obviously $\succeq^*$ and $\succeq^{**}$ are different.

Question 1.D.2. Show that if X is finite, then any rational preference relation generates a nonempty choice rule; that is, $C(B) \neq \emptyset$ for any $B \subset X$ with $B \neq \emptyset$.

Answer 1.D.2. Suppose $\emptyset \neq B \in \mathcal{B}$. If B has one element, by definition, $C(B, \succeq) = B \neq \emptyset$. Now, suppose that B has $n > 1$ elements $x_1, \dots, x_n$. Since $\succeq$ is complete, $x_i \succeq x_j$ or $x_j \succeq x_i$ for all $i, j = 1, \dots, n$. Since $\succeq$ is transitive, we can apply the pairwise preferences to find a permutation σ on $1, \dots, n$ such that $x_{\sigma(1)} \succeq x_{\sigma(2)} \succeq \dots \succeq x_{\sigma(n)}$. Thus, $x_{\sigma(1)} \in C(B, \succeq)$, so $C(B, \succeq)$ is not empty.

Question 1.D.3. Let $X = \{x, y, z\}$, and consider the choice structure $(\mathcal{B}, C(\cdot))$ with

$$\mathcal{B} = \{\{x, y\}, \{y, z\}, \{x, z\}, \{x, y, z\}\}$$

and $C(\{x, y\}) = \{x\}$, $C(\{y, z\}) = \{y\}$, and $C(\{x, z\}) = \{z\}$, as in Example 1.D.1. Show that $(\mathcal{B}, C(\cdot))$ must violate the weak axiom.

Answer 1.D.3. Suppose the weak axiom holds. We will show that this assumption implies that $C(\{x, y, z\})$ is not well-defined.

First, if the weak axiom holds, $C(\{x, y, z\})$ must contain x since $\{x, y\} \subset \{x, y, z\}$ and $C(\{x, y\}) = \{x\}$. However, $C(\{x, y, z\})$ also must not contain x , since $\{x, z\} \subset \{x, y, z\}$ and $\{x\} \not\subset C(\{x, z\})$. Since the weak axiom implies $x \in C(\{x, y, z\})$ and $x \notin C(\{x, y, z\})$, the weak axiom must not apply to this choice structure.

Question 1.D.4. Show that a choice structure $(\mathcal{B}, C(\cdot))$ for which a rationalizing preference relation $\succeq$ exists satisfies the *path-invariance* property: For every pair $B_1, B_2 \in \mathcal{B}$ such that $B_1 \cup B_2 \in \mathcal{B}$ and $C(B_1) \cup C(B_2) \in \mathcal{B}$, we have $C(B_1 \cup B_2) = C(C(B_1) \cup C(B_2))$, that is, the decision problem can be safely subdivided. See Plott (1973) for further discussion.

Answer 1.D.4. First, show that $C(B_1 \cup B_2) \subseteq C(C(B_1) \cup C(B_2))$. Suppose $x \in C(B_1 \cup B_2)$. Then $x \succeq y$ for all $y \in B_1 \cup B_2$. Since $B_1 \subseteq B_1 \cup B_2$ and $B_2 \subseteq B_1 \cup B_2$,

- $x \succeq y$ for all $y \in B_1$, which implies $x \in C(B_1)$
- $x \succeq y$ for all $y \in B_2$, which implies $x \in C(B_2)$

Thus, $x \in C(B_1) \cup C(B_2) \subseteq C(B_1 \cup B_2)$, so $x \succeq y$ for every $y \in C(B_1) \cup C(B_2)$, i.e. $x \in C(C(B_1) \cup C(B_2))$. Therefore, $C(B_1 \cup B_2) \subseteq C(C(B_1) \cup C(B_2))$.

Second, show that $C(C(B_1) \cup C(B_2)) \subseteq C(B_1 \cup B_2)$. Suppose $x \in C(C(B_1) \cup C(B_2))$. Then $x \succeq y$ for all $y \in C(B_1) \cup C(B_2)$. Since $C(B_1) \subseteq C(B_1) \cup C(B_2)$ and $C(B_2) \subseteq C(B_1) \cup C(B_2)$,

- $x \succeq y$ for all $y \in C(B_1) = \{y \in B_1 | y \succeq z \text{ for all } z \in B_1\}$
- $x \succeq y$ for all $y \in C(B_2) = \{y \in B_2 | y \succeq z \text{ for all } z \in B_2\}$

By transitivity, $x \succeq z$ for all $z \in B_1 \cup B_2$, i.e. $x \in C(B_1 \cup B_2)$. Therefore, $C(C(B_1) \cup C(B_2)) \subseteq C(B_1 \cup B_2)$.

Since $C(B_1 \cup B_2) \subseteq C(C(B_1) \cup C(B_2))$ and $C(C(B_1) \cup C(B_2)) \subseteq C(B_1 \cup B_2)$, $C(B_1 \cup B_2) = C(C(B_1) \cup C(B_2))$.

Question 1.D.5. Let $X = \{x, y, z\}$ and $\mathcal{B} = \{\{x, y\}, \{y, z\}, \{x, z\}\}$. Suppose that choice is now stochastic in the sense that for every $B \in \mathcal{B}$, $C(B)$ is a frequency distribution over alternatives in B . For example, if $B = \{x, y\}$, we write $C(B) = (C_x(B), C_y(B))$, where $C_x(B)$ and $C_y(B)$ are nonnegative numbers with $C_x(B) + C_y(B) = 1$. We say that the stochastic choice function $C(\cdot)$ can be *rationalized by preferences* if we can find a probability distribution Pr over the six possible (strict) preference relations on X such that for every $B \in \mathcal{B}$, $C(B)$ is precisely the frequency of choices induced by Pr . For example, if $B = \{x, y\}$, then $C_x(B) = Pr(\{ \succ | x \succ y \})$.

This concept originates in Thurstone (1927), and it is of considerable econometric interest (indeed, it provides a theory for the error term in observable choice).

Question 1.D.5.a. Show that the stochastic choice function $C(\{x, y\}) = C(\{y, z\}) = C(\{x, z\}) = (\frac{1}{2}, \frac{1}{2})$ can be rationalized by preferences.

Answer 1.D.5.a. Define

$$Pr : \begin{cases} x \succ y \succ z \rightarrow 1/6 \\ x \succ z \succ y \rightarrow 1/6 \\ y \succ x \succ z \rightarrow 1/6 \\ y \succ z \succ x \rightarrow 1/6 \\ z \succ x \succ y \rightarrow 1/6 \\ z \succ y \succ x \rightarrow 1/6 \end{cases}$$

Then we have

$$\begin{aligned} \left. \begin{aligned} Pr(x \succ y) &= 1/2 \\ Pr(y \succ x) &= 1/2 \end{aligned} \right\} & \text{implies } C(\{x, y\}) = (1/2, 1/2) \\ \left. \begin{aligned} Pr(y \succ z) &= \frac{1}{2} \\ Pr(z \succ y) &= \frac{1}{2} \end{aligned} \right\} & \text{implies } C(\{y, z\}) = (1/2, 1/2) \\ \left. \begin{aligned} Pr(x \succ z) &= \frac{1}{2} \\ Pr(z \succ x) &= \frac{1}{2} \end{aligned} \right\} & \text{implies } C(\{x, z\}) = (1/2, 1/2) \end{aligned}$$

So C can be rationalized by preferences.

Question 1.D.5.b. Show that the stochastic choice function $C(\{x, y\}) = C(\{y, z\}) = C(\{x, z\}) = (\frac{1}{4}, \frac{3}{4})$ is not rationalizable by preferences.

Answer 1.D.5.b. Define

$$Pr : \begin{cases} x \succ y \succ z \rightarrow p_1 \\ x \succ z \succ y \rightarrow p_2 \\ y \succ x \succ z \rightarrow p_3 \\ y \succ z \succ x \rightarrow p_4 \\ z \succ x \succ y \rightarrow p_5 \\ z \succ y \succ x \rightarrow p_6 \end{cases}$$

If C can be rationalized by preferences then the system

$$p_1 + p_2 + p_5 = 1/4 \quad (1)$$

$$p_3 + p_4 + p_6 = 3/4$$

$$p_1 + p_3 + p_4 = 1/4 \quad (2)$$

$$p_2 + p_5 + p_6 = 3/4$$

$$p_4 + p_5 + p_6 = 1/4$$

$$p_1 + p_2 + p_3 = 3/4 \quad (3)$$

has a solution with $p_i \in [0, 1]$ for $i = 1, \dots, 6$. However, Equation (1) implies $p_1 + p_2 \leq 1/4$. Substituting this inequality into Equation (3), $p_3 \geq 1/2$. However, then Equation (2) implies that $p_1 + p_4 \leq -1/4$, contradicting $p_i \in [0, 1]$ for $i = 1, \dots, 6$. Therefore, C cannot be rationalized by preferences.

Question 1.D.5.c. Determine the $0 < \alpha < 1$ at which $C(\{x, y\}) = C(\{y, z\}) = C(\{x, z\}) = (\alpha, 1 - \alpha)$ switches from rationalizable to non rationalizable.

Answer 1.D.5.c. From Exercise 1.D.5.b, it follows that C is rationalizable exactly when α is such that the system

$$p_1 + p_2 + p_5 = \alpha$$

$$p_3 + p_4 + p_6 = 1 - \alpha$$

$$p_1 + p_3 + p_4 = \alpha$$

$$p_2 + p_5 + p_6 = 1 - \alpha$$

$$p_4 + p_5 + p_6 = \alpha$$

$$p_1 + p_2 + p_3 = 1 - \alpha$$

has a solution with $p_i \in [0, 1]$ for $i = 1, \dots, 6$. Generalizing from Exercise 1.D.5.b, we require

$$p_1 + p_2 \leq \alpha \Rightarrow p_3 \geq 1 - 2\alpha \Rightarrow p_1 + p_4 \leq 3\alpha - 1$$

which suggests that the choice structure is rationalizable as long as $3\alpha - 1 \geq 0$, or $\alpha \geq 1/3$. However, by symmetry, we also require $1 - \alpha \geq 1/3$, so C is rationalizable by preferences whenever $1/3 \leq \alpha \leq 2/3$.

Chapter 2

Question 2.D.1. A consumer lives for two periods, denoted 1 and 2, and consumes a single consumption good in each period. His wealth when born is $w > 0$. What is his (lifetime) Walrasian budget set?

Answer 2.D.1. The consumer's Walrasian budget set is $\{(c_1, c_2) \in \mathbb{R}_+^2 | p_1 c_1 + p_2 c_2 \leq w\}$ where w is lifetime wealth, p_1 is the price of the good in period 1, and p_2 is the price of the good in period 2.

“Official” solutions normalize $p_1 = 1$.

Question 2.D.2. A consumer consumes one consumption good x and hours of leisure h . The price of the consumption good is p , and the consumer can work at a wage rate of $s = 1$. What is the consumer's Walrasian budget set?

Answer 2.D.2. The consumer's wealth is $w = (24 - h)s = 24 - h$ which he can spend on good x at price p . The consumer's Walrasian budget set is $\{(x, h) \in \mathbb{R}_+^2 | px \leq 24 - h\}$.

“Official” solutions normalize $\{(x, h) \in \mathbb{R}_+^2 | px + h \leq 24\}$.

Question 2.D.3. Consider an extension of the Walrasian budget set to an arbitrary consumption set $X : B_{p,w} = \{x \in X | p \cdot x \leq w\}$. Assume $(p, w) \gg 0$.

Question 2.D.3.a. If X is the set depicted in Figure 2.C.3, would $B_{p,w}$ be convex?

Answer 2.D.3.a. No, since $B_{p,w}$ contains $(0, w/p_2)$ and $(w/p_1, 0)$ but does not contain $(\alpha w/p_1, (1 - \alpha)w/p_2)$ unless $\alpha = 0$ or 1 .

Question 2.D.3.b. Show that if X is a convex set, $B_{p,w}$ is as well.

Answer 2.D.3.b. Let x and y be any elements of $B_{p,w}$ and let $\alpha \in (0, 1)$. Since X is convex, $\alpha x + (1 - \alpha)y \in X$. Further, since $x, y \in B_{p,w}$:

$$\begin{aligned} p \cdot (\alpha x + (1 - \alpha)y) &= p \cdot \alpha x + p \cdot (1 - \alpha)y \\ p \cdot (\alpha x + (1 - \alpha)y) &\leq \alpha w + (1 - \alpha)w \text{ since } x, y \in B_{p,w} \\ p \cdot (\alpha x + (1 - \alpha)y) &\leq w \end{aligned}$$

Therefore, since $\alpha x + (1 - \alpha)y \in X$ and $p \cdot (\alpha x + (1 - \alpha)y) \leq w$, it follows that $\alpha x + (1 - \alpha)y \in B_{p,w}$, i.e. $B_{p,w}$ is convex.

Question 2.D.4. Show that the budget set in Figure 2.D.4 is not convex.

Answer 2.D.4. Denote Leisure Hours by h and Consumption Good by g . The budget set in Figure 2.D.4 is the region

$$\left\{ (h, g) \in \mathbb{R}_+^2 | g \leq 24 - sh, g \leq -s'(h - 16) + 8s, g \leq -s'(1 - t) \left(h - \left(16 - \frac{M - 8s}{s'} \right) \right) + M \right\}$$

This region is not convex because it contains the points $(24, 0)$ and $(16 - \frac{M-8s}{s'}, M)$ but does not contain the point

$$\left(1 - \frac{8s}{M}\right)(24, 0) + \frac{8s}{M}\left(16 - \frac{M-8s}{s'}, M\right)$$

which is a convex combination since $0 < 8s < M$ implies $0 < \frac{8s}{M} < 1$.

Question 2.E.1. Suppose $L = 3$ and consider the demand function $x(p, w)$ defined by

$$\begin{aligned} x_1(p, w) &= \frac{p_2}{p_1 + p_2 + p_3} \frac{w}{p_1} \\ x_2(p, w) &= \frac{p_3}{p_1 + p_2 + p_3} \frac{w}{p_2} \\ x_3(p, w) &= \frac{\beta p_1}{p_1 + p_2 + p_3} \frac{w}{p_3} \end{aligned}$$

Does this function satisfy homogeneity of degree zero and Walras' law when $\beta = 1$? What about when $\beta \in (0, 1)$?

Answer 2.E.1. When $\beta = 1$, $x(p, w)$ satisfies homogeneity of degree zero since for $\alpha > 0$

$$x_1(\alpha p, \alpha w) = \frac{\alpha p_2}{\alpha(p_1 + p_2 + p_3)} \frac{\alpha w}{\alpha p_1} = \frac{p_2}{p_1 + p_2 + p_3} \frac{w}{p_1} = x_1(p, w)$$

and similarly for $x_2(p, w)$ and $x_3(p, w)$.

When $\beta = 1$, $x(p, w)$ satisfies Walras' law since:

$$p_1 x_1(p, w) + p_2 x_2(p, w) + p_3 x_3(p, w) = \frac{p_2 w + p_3 w + p_1 w}{p_1 + p_2 + p_3} = w$$

When $\beta \in (0, 1)$ homogeneity of degree zero still holds but Walras' law fails.

Question 2.E.2. Show that equations (2.E.4) and (2.E.6) lead to the following two elasticity formulas:

$$\sum_{l=1}^L b_l(p, w) \epsilon_{lk}(p, w) + b_k(p, w) = 0$$

and

$$\sum_{l=1}^L b_l(p, w) \epsilon_{lw}(p, w) = 1$$

where $b_l(p, w) = p_l x_l(p, w)/w$ is the budget share of the consumer's expenditure on good l given prices p and wealth w .

Answer 2.E.2. Recall that

$$\epsilon_{lk}(p, w) = \frac{\partial x_l(p, w)}{\partial p_k} \frac{p_k}{x_l(p, w)} \text{ and } \epsilon_{lw}(p, w) = \frac{\partial x_l(p, w)}{\partial w} \frac{w}{x_l(p, w)}$$

Equation 2.E.4 from the text is

$$0 = \sum_{l=1}^L p_l \frac{\partial x_l(p, w)}{\partial p_k} + x_k(p, w)$$

Substituting $\frac{\partial x_l(p, w)}{\partial p_k} = \epsilon_{lk}(p, w) \frac{x_l(p, w)}{p_k}$:

$$0 = \sum_{l=1}^L p_l \epsilon_{lk}(p, w) \frac{x_l(p, w)}{p_k} + x_k(p, w)$$

Substituting $w b_l(p, w) = x_l(p, w) p_l$:

$$0 = \sum_{l=1}^L \epsilon_{lk}(p, w) \frac{w b_l(p, w)}{p_k} + w b_k(p, w) / p_k$$

Multiplying through by p_k/w :

$$0 = \sum_{l=1}^L b_l(p, w) \epsilon_{lk}(p, w) + b_k(p, w)$$

Equation 2.E.6 from the text is

$$1 = \sum_{l=1}^L p_l \frac{\partial x_l(p, w)}{\partial w}$$

Substituting $\frac{\partial x_l(p, w)}{\partial w} = \epsilon_{lw}(p, w) \frac{x_l(p, w)}{w}$:

$$1 = \sum_{l=1}^L p_l x_l(p, w) \epsilon_{lw}(p, w) / w$$

Substituting $b_l(p, w) = p_l x_l(p, w) / w$:

$$1 = \sum_{l=1}^L b_l(p, w) \epsilon_{lw}(p, w)$$

Question 2.E.3. Use Propositions 2.E.1 and 2.E.3 to show that $p \cdot D_p x(p, w) p = -w$. Interpret.

Answer 2.E.3. Proposition 2.E.1 states that

$$\begin{aligned} D_p x(p, w)p + D_w x(p, w)w &= 0 \\ p \cdot D_p x(p, w)p + p \cdot D_w x(p, w)w &= 0 \end{aligned}$$

Substituting $p \cdot D_w x(p, w) = 1$ (Proposition 2.E.3):

$$\begin{aligned} p \cdot D_p x(p, w)p + w &= 0 \\ p \cdot D_p x(p, w)p &= -w \end{aligned}$$

“Official solutions” interpret this to mean that $x(p, w)$ is homogeneous of degree 0 which was an assumption of Proposition 2.E.1 to begin with.

Question 2.E.4. This question was on Emel Filiz-Ozbay’s August 2009 Comp. Show that if $x(p, w)$ is homogeneous of degree one with respect to w [i.e., $x(p, \alpha w) = \alpha x(p, w)$ for all $\alpha > 0$] and satisfies Walras’ law, then $\epsilon_{lw}(p, w) = 1$ for every l . Interpret. Can you say something about $D_w x(p, w)$ and the form of the Engel functions and curves in this case?

Answer 2.E.4. Solution 1 (From Miguel Almanzar):

Since $x(p, w)$ is homogeneous of degree 0, it must be that $x_l(p, w)$ is homogeneous of degree 0 for every l , i.e.

$$x_l(p, \alpha w) = \alpha x_l(p, w) \text{ for all } \alpha > 0$$

In particular, for $\alpha = w^{-1}$:

$$wx_l(p, 1) = x_l(p, w)$$

Differentiating both sides with respect to w :

$$\begin{aligned} x_l(p, 1) &= \frac{\partial x_l(p, w)}{\partial w} \\ x_l(p, 1) \frac{w}{x_l(p, w)} &= \frac{\partial x_l(p, w)}{\partial w} \frac{w}{x_l(p, w)} \\ x_l(p, 1) \frac{w}{wx_l(p, 1)} &= \frac{\partial x_l(p, w)}{\partial w} \frac{w}{x_l(p, w)} \text{ because } x_l(p, w) \text{ is homogeneous of degree 0} \\ 1 &= \frac{\partial x_l(p, w)}{\partial w} \frac{w}{x_l(p, w)} \\ 1 &= \epsilon_{lw}(p, w) \end{aligned}$$

Solution 2 (From Sinem Kilic):

Since $x(p, w)$ is homogeneous of degree 0, it must be that $x_l(p, w)$ is homogeneous of degree 0 for every l , i.e.

$$x_l(p, \alpha w) = \alpha x_l(p, w) \text{ for all } \alpha > 0$$

Differentiate with respect to α and evaluate at $\alpha = 1$:

$$\begin{aligned} w \frac{\partial x_l(p, w)}{\partial w} &= x_l(p, w) \\ 1 &= \frac{\partial x_l(p, w)}{\partial w} \frac{x_l(p, w)}{w} \\ 1 &= \epsilon_{lw}(p, w) \end{aligned}$$

Interpretation: For any fixed price vector p , each good's budget share is constant, i.e. the Engel functions / curves are straight lines.

Question 2.E.5. [This question was on Emel Filiz-Ozbay's August 2009 Comp.](#) Suppose that $x(p, w)$ is a demand function which is homogeneous of degree 1 with respect to w and satisfies Walras' law and homogeneity of degree 0. Suppose also that all the cross-price effects are zero, that is $\partial x_l(p, w) / \partial p_k = 0$ whenever $k \neq l$. Show that this implies that for every l , $x_l(p, w) = \alpha_l w / p_l$, where $\alpha_l > 0$ is a constant independent of (p, w) .

Answer 2.E.5. Since $x(p, w)$ is homogeneous of degree 1 with respect to w , $x(p, w) = \frac{w}{p_l} x(p, p_l)$. Therefore, in particular, $x_l(p, w) = \frac{w}{p_l} x_l(p, p_l)$. Put $\alpha_l = x_l(p, p_l)$. Now we show that α_l is independent of (p, w) . Obviously, α_l is independent of w . To see that α_l is independent of prices:

$$\frac{\partial \alpha_l}{\partial p_k} = \frac{\partial}{\partial p_k} x_l(p, p_l)$$

If $k \neq l$,

$$\frac{\partial \alpha_l}{\partial p_k} = 0$$

because all cross price effects are 0. If $k = l$:

$$\frac{\partial \alpha_l}{\partial p_l} = \frac{\partial}{\partial p_l} x_l(p, p_l)$$

By the chain rule:

$$\begin{aligned}
 &= \frac{\partial x_l(p, p_l)}{\partial p_l} + \frac{\partial x_l(p, p_l)}{\partial w} \frac{\partial p_l}{\partial p_l} \\
 &= \frac{\partial x_l(p, p_l)}{\partial p_l} + \frac{\partial x_l(p, p_l)}{\partial w} \\
 &= \frac{1}{p_l} \left(\frac{\partial x_l(p, p_l)}{\partial p_l} p_l + \frac{\partial x_l(p, p_l)}{\partial w} p_l \right)
 \end{aligned}$$

By Proposition 2.E.1 (which requires homogeneity of degree 0)

$$= 0$$

Therefore, α_l is independent of (p, w) .

Last but not least, we show by contradiction that $\alpha_l > 0$. Obviously $\alpha_l \geq 0$. Suppose $\alpha_l = 0$. Then since $x(p, w)$ is homogeneous of degree 1 in w , it follows from $x_l(p, w) = \alpha_l w/p_l$ that $x_l(p, w) = 0$ for all (p, w) . In other words, we can just eliminate good l from consideration, so it was never part of the problem to begin with.

From Grace Noboa: Note that we did *not* use Walras' Law. We do not need it, and in fact the "official" solutions do not use it either.

Question 2.E.6. Verify that the conclusions of Propositions 2.E.1 to 2.E.3 hold for the demand function given in Exercise 2.E.1 when $\beta = 1$.

Answer 2.E.6. From Exercise 2.E.1, when $\beta = 1$, the demand function satisfies Walras' law and homogeneity of degree 0, which are the hypotheses of Propositions 2.E.1 to 2.E.3, therefore these propositions hold.

Question 2.E.7. A consumer in a two-good economy has a demand function $x(p, w)$ that satisfies Walras' law. His demand function for the first good is $x_1(p, w) = \alpha w/p_1$. Derive his demand function for the second good. Is his demand function homogeneous of degree zero?

Answer 2.E.7. By Walras' law, $p_1(\alpha w/p_1) + p_2 x_2 = w$, which immediately implies that $x_2(p, w) = (1 - \alpha)w/p_2$. His demand is homogeneous of degree 0 since for all $\beta > 0$

$$\begin{aligned}
 x(\beta p, \beta w) &= (\alpha \beta w / (\beta p_1), (1 - \alpha) \beta w / (\beta p_2)) \\
 &= (\alpha w / p_1, (1 - \alpha) w / p_2) \\
 &= x(p, w)
 \end{aligned}$$

Question 2.E.8. Show that the elasticity of demand for good l with respect to price p_k , $\epsilon_{lk}(p, w)$, can be written as $\epsilon_{lk}(p, w) = d \ln x_l(p, w) / d \ln(p_k)$, where $\ln(\cdot)$ is the natural logarithm function. Derive a similar expression for $\epsilon_{lw}(p, w)$. Conclude that if we estimate the parameters $(\alpha_0, \alpha_1, \alpha_2, \gamma)$ of the equation $\ln(x_l(p, w)) = \alpha_0 + \alpha_1 \ln p_1 + \alpha_2 \ln p_2 + \gamma \ln w$, these parameter estimates provide us with estimates of the elasticities $\epsilon_{l1}(p, w)$, $\epsilon_{l2}(p, w)$, and $\epsilon_{lw}(p, w)$.

Answer 2.E.8. By definition:

$$\epsilon_{lk}(p, w) = \frac{\partial x_l(p, w)}{\partial p_k} \frac{p_k}{x_l(p, w)}$$

Solving the differential equation on the right-hand side:

$$\epsilon_{lk}(p, w) = \frac{\partial \ln x_l(p, w)}{\partial \ln p_k}$$

By definition:

$$\epsilon_{lw}(p, w) = \frac{\partial x_l(p, w)}{\partial w} \frac{w}{x_l(p, w)}$$

Solving the differential equation on the right-hand side:

$$\epsilon_{lw}(p, w) = \frac{\partial \ln x_l(p, w)}{\partial \ln w}$$

Assuming we have an estimate for $\ln(x_l(p, w)) = \alpha_0 + \alpha_1 \ln p_1 + \alpha_2 \ln p_2 + \gamma \ln w$,

$$\begin{aligned} \frac{\partial \ln x_l(p, w)}{\partial \ln p_1} &= \alpha_1 \\ \frac{\partial \ln x_l(p, w)}{\partial \ln p_2} &= \alpha_2 \\ \frac{\partial \ln x_l(p, w)}{\partial \ln w} &= \gamma \end{aligned}$$

Question 2.F.1. Show that for Walrasian demand functions, the definition of the weak axiom given in Definition 2.F.1 coincides with that in Definition 1.C.1.

Answer 2.F.1. We restate these definitions for convenience:

Definition 1.C.1: The choice structure $(\mathcal{B}, C(\cdot))$ satisfies the weak axiom of revealed preference if for some $B \in \mathcal{B}$ with $x, y \in B$ we have $x \in C(B)$, then for any $B' \in \mathcal{B}$ with $x, y \in B'$ and $y \in C(B')$, we must also have $x \in C(B')$.

Definition 2.F.1: The Walrasian demand function $x(p, w)$ satisfies the weak axiom of revealed preference if for any two price-wealth situations (p, w) and (p', w') , if $p \cdot x(p', w') \leq w$ and $x(p', w') \neq x(p, w)$, then $p' \cdot x(p, w) > w'$.

Let $B = \{(x_1, \dots, x_L) \in \mathbb{R}_+ | p \cdot (x_1, \dots, x_L) \leq w\}$ and let $B' = \{(x_1, \dots, x_L) \in \mathbb{R}_+ | p' \cdot (x_1, \dots, x_L) \leq w'\}$. Let the choice structure be exactly the Walrasian demand function. Then

By Walras' law

$$\begin{aligned} x(p, w) &\in B, C(B) \\ x(p', w') &\in B', C(B') \end{aligned}$$

By construction

$$\begin{aligned} p \cdot x(p', w') &\leq w \text{ is equivalent to } x(p', w') \in B \\ x(p', w') &\neq x(p, w) \text{ is equivalent to } x(p', w') \notin C(B), x(p, w) \notin C(B') \\ p' \cdot x(p, w) &> w' \text{ is equivalent to } x(p, w) \notin B' \end{aligned}$$

Rewriting Definition 2.F.1 in logically equivalent terms: If $x(p, w), x(p', w') \in B$ and $x(p, w) \in C(B)$, then for any B' with $x(p', w') \in C(B')$, if $x(p, w) \notin C(B')$ then $x(p, w) \notin B'$. Since “if $x(p, w) \notin C(B')$ then $x(p, w) \notin B'$ ” is the contrapositive of “if $x(p, w) \in B'$ then $x(p, w) \in C(B')$,” which is exactly the concluding statement in Definition 1.C.1, we are done.

Question 2.F.2. This question was on Emel Filiz-Ozbay's Fall 2008 Midterm. Verify the claim of Example 2.F.1.

Answer 2.F.2.

p	w	Choice
$p^1 = (2, 1, 2)$	8	$x^1 = (1, 2, 2)$
$p^2 = (2, 2, 1)$	8	$x^2 = (2, 1, 2)$
$p^3 = (1, 2, 2)$	8	$x^3 = (2, 2, 1)$

The pair x^1 and x^2 satisfy WARP since $p^2 \cdot x^1 \leq 8$, $x^1 \neq x^2$ and $p^1 \cdot x^2 = 9 > 8$. Therefore, $x^2 \succeq_{rp} x^1$. Similarly, $x^3 \succeq_{rp} x^2$ and $x^1 \succeq_{rp} x^3$. Since the revealed preference relation violates transitivity, it is not rational.

Question 2.F.3. This question was on Emel Filiz-Ozbay's Fall 2009 Midterm. You are given the following partial information about a consumer's purchases. He

consumes only two goods.

	Year 1		Year 2	
	Quantity	Price	Quantity	Price
Good 1	100	100	120	100
Good 2	100	100	y	80

Over what range of y would you conclude:

Question 2.F.3.a. That his behavior is inconsistent (i.e., in contradiction with the weak axiom)?

Answer 2.F.3.a. Find y such that both bundles are affordable in both years, i.e.

$$100 * 120 + 100 * y \leq 100 * 100 + 100 * 100$$

$$100 * 100 + 80 * 100 \leq 100 * 120 + 80 * y$$

Simplifying these inequalities, the weak axiom is violated when

$$75 \leq y \leq 80$$

Question 2.F.3.b. That the consumer's consumption bundle in year 1 is revealed preferred to that in year 2?

Answer 2.F.3.b. Find y such that

$$100 * 120 + 100 * y \leq 100 * 100 + 100 * 100$$

$$100 * 100 + 80 * 100 > 100 * 120 + 80 * y \text{ (to additionally be consistent with WARP)}$$

Simplifying these inequalities, the consumer's consumption bundle in year 1 is revealed preferred to that in year 2

$$y \leq 80, (y < 75 \text{ to additionally be consistent with WARP})$$

Question 2.F.3.c. That the consumer's consumption bundle in year 2 is revealed preferred to that in year 1?

Answer 2.F.3.c. Reversing the inequalities above, $y \geq 75$, and $y > 80$ to additionally be consistent with WARP.

Question 2.F.3.d. That there is insufficient information to justify (a), (b), or (c)?

Answer 2.F.3.d. Any value of y will satisfy at least one of the above sets of inequalities, so there will always be enough information to justify at least one of (a), (b), or (c).

Question 2.F.3.e. That good 1 is an inferior good (at some price) for this consumer? Assume that the weak axiom is satisfied.

Answer 2.F.3.e. We need to decompose the change in demand from Year 2 to Year 1 into price effect and wealth effect. The good 2 consumption level at which the price change from Year 2 to Year 1 is compensated is when the Year 2 bundle is exactly affordable at Year 1 prices and wealth. The wealth level to compensate the price change from Year 1 to Year 2 is cost of the Year 1 bundle at Year 2 prices

$$100 * 100 + 100 * 80 = 18000$$

The consumption level in Year 2 at that wealth level satisfies

$$\begin{aligned} 120 * 100 + y * 80 &= 18000 \\ y &= 75 \end{aligned}$$

Therefore, real wealth decreased from year 1 to year 2 if $y < 75$. If good 1 is inferior, we require that $y < 75$, so that while real wealth decreased, consumption of good 1 increased. Therefore, good 1 is inferior at some price if $y < 75$. From Exercise 2.F.3.a, we know that the weak axiom is also satisfied for $y < 75$.

Question 2.F.3.f. That good 2 is an inferior good (at some price) for this consumer? Assume that the weak axiom is satisfied.

Answer 2.F.3.f. We need to decompose the change in demand from Year 1 to Year 2 into price effect and wealth effect. The wealth level to compensate the price change from Year 1 to Year 2 is cost of the Year 1 bundle at Year 2 prices

$$100 * 100 + 100 * 80 = 18000$$

The consumption level in Year 2 at that wealth level satisfies

$$\begin{aligned} 120 * 100 + y * 80 &= 18000 \\ y &= 75 \end{aligned}$$

Therefore, real wealth increased from year 1 to year 2 if $y > 75$. If good 2 is inferior, we require that $y < 100$, so that while real wealth increased, consumption of good 2 decreased. Therefore, good 2 is inferior at some price if $75 < y < 100$. Since we further require that the weak axiom hold, good 2 is inferior when $80 < y < 100$.

Question 2.F.4. Consider the consumption of a consumer in two different periods, period 0 and period 1. Period t prices, wealth, and consumption are p^t, w_t , and $x^t = x(p^t, w_t)$, respectively. It is often of applied interest to form an index measure of the quantity consumed by a consumer. The *Laspeyres* quantity index computes the change in quantity using period 0 prices as weights: $L_Q = (p^0 \cdot x^1)/(p^0 \cdot x^0)$. The *Paasche* quantity index instead uses period 1 prices as weights: $P_Q = (p^1 \cdot x^1)/(p^1 \cdot x^0)$. Finally, we could use the consumer's expenditure change: $E_Q = (p^1 \cdot x^1)/(p^0 \cdot x^0)$. Show the following:

Question 2.F.4.a. If $L_Q < 1$, then the consumer has a revealed preference for x^0 over x^1 .

Answer 2.F.4.a. If $L_Q < 1$, then $p^0 \cdot x^1 < p^0 \cdot x^0 = w_0$, i.e. the consumer can afford both x^0 and x^1 given p^0, w_0 and chose x^0 .

Question 2.F.4.b. If $P_Q > 1$, then the consumer has a revealed preference for x^1 over x^0 .

Answer 2.F.4.b. If $P_Q > 1$, then $p^1 \cdot x^0 < p^1 \cdot x^1 = w_1$, i.e. the consumer can afford both x^0 and x^1 given p^1, w_1 and chose x^1 .

Question 2.F.4.c. No revealed preference relationship is implied by either $E_Q > 1$ or $E_Q < 1$. Note that at the aggregate level, E_Q corresponds to the percentage change in gross national product.

Answer 2.F.4.c. The relations $E_Q > 1$ or $E_Q < 1$ don't tell us anything that compares consumption at a single price-wealth combination, so it can't imply anything about revealed preference.

Question 2.F.5. Suppose that $x(p, w)$ is a differentiable demand function that satisfies the weak axiom, Walras' law, and homogeneity of degree zero. Show that if $x(\cdot, \cdot)$ is homogeneous of degree one with respect to w [i.e. $x(p, \alpha w) = \alpha x(p, w)$ for all (p, w) and $\alpha > 0$], then the law of demand holds even for *uncompensated* price changes. If this is easier, establish only the infinitesimal version of this conclusion; that is $dp \cdot D_p x(p, w) dp \leq 0$ for any dp .

Answer 2.F.5.

Question 2.F.7. Prove Proposition 2.F.3. [*Hint:* Use Propositions 2.E.1 to 2.E.3.]

Answer 2.F.7. First, show that $p \cdot S(p, w) = 0$. Expanding

$$p \cdot S(p, w) = \begin{bmatrix} \sum_{l=1}^L p_l s_{l1} \\ \vdots \\ \sum_{l=1}^L p_l s_{lL} \end{bmatrix}$$

Consider the k th entry:

$$\begin{aligned} \sum_{l=1}^L p_l s_{lk} &= \sum_{l=1}^L p_l \left[\frac{\partial x_l(p, w)}{\partial p_k} + \frac{\partial x_l(p, w)}{\partial w} x_k(p, w) \right] \\ &= \underbrace{\sum_{l=1}^L \frac{\partial x_l(p, w)}{\partial p_k} p_l}_{=-x_k(p, w) \text{ by Proposition 2.E.2}} + x_k(p, w) \underbrace{\sum_{l=1}^L \frac{\partial x_l(p, w)}{\partial w} p_l}_{=1 \text{ by Proposition 2.E.3}} \\ &= 0 \end{aligned}$$

Now, show that $S(p, w)p = 0$. Expanding

$$S(p, w)p = \begin{bmatrix} \sum_{k=1}^L p_k s_{1k} \\ \vdots \\ \sum_{k=1}^L p_k s_{Lk} \end{bmatrix}$$

Consider the l th entry:

$$\begin{aligned} \sum_{k=1}^L p_k s_{lk} &= \sum_{k=1}^L p_k \left[\frac{\partial x_l(p, w)}{\partial p_k} + \frac{\partial x_l(p, w)}{\partial w} x_k(p, w) \right] \\ &= \underbrace{\sum_{k=1}^L \frac{\partial x_l(p, w)}{\partial p_k} p_k}_{=-\frac{\partial x_l(p, w)}{\partial w} w \text{ by Proposition 2.E.1}} + \frac{\partial x_l(p, w)}{\partial w} \underbrace{\sum_{k=1}^L x_k(p, w) p_k}_{=1 \text{ by Walras' law}} \\ &= 0 \end{aligned}$$

Question 2.F.8. Let $\hat{s}_{lk}(p, w) = [p_k/x_l(p, w)]s_{lk}(p, w)$ be the substitution terms in elasticity form. Express $\hat{s}_{lk}(p, w)$ in terms of $\epsilon_{lk}(p, w)$, $\epsilon_l w(p, w)$, and $b_k(p, w)$.

Answer 2.F.8. Recall the definition of $b_k(p, w)$ from Exercise 2.E.2.

$$\begin{aligned}
 \hat{s}_{lk}(p, w) &= [p_k/x_l(p, w)]s_{lk}(p, w) \\
 &= \frac{p_k}{x_l(p, w)} \left[\frac{\partial x_l(p, w)}{\partial p_k} + \frac{\partial x_l(p, w)}{\partial w} x_k(p, w) \right] \\
 &= \frac{\partial x_l(p, w)}{\partial p_k} \frac{p_k}{x_l(p, w)} + \frac{\partial x_l(p, w)}{\partial w} \frac{p_k x_k(p, w)}{x_l(p, w)} \\
 &= \epsilon_{lk}(p, w) + \frac{\partial x_l(p, w)}{\partial w} \frac{w}{x_l(p, w)} \frac{p_k x_k(p, w)}{w} \\
 &= \epsilon_{lk}(p, w) + \epsilon_{lw}(p, w) b_k(p, w)
 \end{aligned}$$

Question 2.F.10. Consider the demand function in Exercise 2.E.1 with $\beta = 1$. Assume that $w = 1$.

Question 2.F.10.a. Compute the Slutsky matrix. Show that at $p = (1, 1, 1)$, it is negative semidefinite but not symmetric.

Answer 2.F.10.a. Recall that the demand function is

$$\begin{aligned}
 x_1(p, w) &= \frac{p_2}{p_1 + p_2 + p_3} \frac{w}{p_1} \\
 x_2(p, w) &= \frac{p_3}{p_1 + p_2 + p_3} \frac{w}{p_2} \\
 x_3(p, w) &= \frac{p_1}{p_1 + p_2 + p_3} \frac{w}{p_3}
 \end{aligned}$$

The lk th entry of the Slutsky matrix is

$$s_{lk} = \frac{\partial x_l(p, w)}{\partial p_l} + \frac{\partial x_l(p, w)}{\partial w} x_k(p, w)$$

Our first step will be to compute all the relevant partial derivatives of the demand functions, then substitute those partial derivatives and evaluate at $p = (1, 1, 1)$,

$w = 1$, to find s_{lk} for all pairs lk .

$$\begin{aligned}
\left. \frac{\partial x_1(p, w)}{\partial p_1} \right|_{p=(1,1,1), w=1} &= - \frac{p_2 w (2p_1 + p_2 + p_3)}{[p_1(p_1 + p_2 + p_3)]^2} \Big|_{p=(1,1,1), w=1} &= -4/9 \\
\left. \frac{\partial x_1(p, w)}{\partial p_2} \right|_{p=(1,1,1), w=1} &= \frac{w(p_1 + p_3)}{p_1(p_1 + p_2 + p_3)^2} \Big|_{p=(1,1,1), w=1} &= 2/9 \\
\left. \frac{\partial x_1(p, w)}{\partial p_3} \right|_{p=(1,1,1), w=1} &= - \frac{p_2 w}{p_1(p_1 + p_2 + p_3)^2} \Big|_{p=(1,1,1), w=1} &= -1/9 \\
\left. \frac{\partial x_1(p, w)}{\partial w} \right|_{p=(1,1,1), w=1} &= \frac{p_2}{p_1(p_1 + p_2 + p_3)} \Big|_{p=(1,1,1), w=1} &= 1/3 \\
\left. \frac{\partial x_2(p, w)}{\partial p_1} \right|_{p=(1,1,1), w=1} &= - \frac{p_3 w}{p_2(p_1 + p_2 + p_3)^2} \Big|_{p=(1,1,1), w=1} &= -1/9 \\
\left. \frac{\partial x_2(p, w)}{\partial p_2} \right|_{p=(1,1,1), w=1} &= - \frac{p_3 w (p_1 + 2p_2 + p_3)}{[p_2(p_1 + p_2 + p_3)]^2} \Big|_{p=(1,1,1), w=1} &= -4/9 \\
\left. \frac{\partial x_2(p, w)}{\partial p_3} \right|_{p=(1,1,1), w=1} &= \frac{w(p_1 + p_2)}{p_2(p_1 + p_2 + p_3)^2} \Big|_{p=(1,1,1), w=1} &= 2/9 \\
\left. \frac{\partial x_2(p, w)}{\partial w} \right|_{p=(1,1,1), w=1} &= \frac{p_3}{p_2(p_1 + p_2 + p_3)} \Big|_{p=(1,1,1), w=1} &= 1/3 \\
\left. \frac{\partial x_3(p, w)}{\partial p_1} \right|_{p=(1,1,1), w=1} &= \frac{w(p_2 + p_3)}{p_3(p_1 + p_2 + p_3)^2} \Big|_{p=(1,1,1), w=1} &= 2/9 \\
\left. \frac{\partial x_3(p, w)}{\partial p_2} \right|_{p=(1,1,1), w=1} &= - \frac{p_1 w}{p_3(p_1 + p_2 + p_3)^3} \Big|_{p=(1,1,1), w=1} &= -1/9 \\
\left. \frac{\partial x_3(p, w)}{\partial p_3} \right|_{p=(1,1,1), w=1} &= - \frac{p_1 w (p_1 + p_2 + 2p_3)}{[p_3(p_1 + p_2 + p_3)]^2} \Big|_{p=(1,1,1), w=1} &= -4/9 \\
\left. \frac{\partial x_3(p, w)}{\partial w} \right|_{p=(1,1,1), w=1} &= \frac{p_1}{p_3(p_1 + p_2 + p_3)} \Big|_{p=(1,1,1), w=1} &= 1/3
\end{aligned}$$

Also, at $p = (1, 1, 1)$ and $w = 1$, the demands are

$$\begin{aligned}
x_1(p, w) &= 1/3 \\
x_2(p, w) &= 1/3 \\
x_3(p, w) &= 1/3
\end{aligned}$$

Therefore the Slutsky matrix is

$$\begin{bmatrix} -4/9 + 1/9 & 2/9 + 1/9 & -1/9 + 1/9 \\ -1/9 + 1/9 & -4/9 + 1/9 & 2/9 + 1/9 \\ 2/9 + 1/9 & -1/9 + 1/9 & -4/9 + 1/9 \end{bmatrix} = \begin{bmatrix} -1/3 & 1/3 & 0 \\ 0 & -1/3 & 1/3 \\ 1/3 & 0 & -1/3 \end{bmatrix}$$

Obviously, this matrix is not symmetric. The eigenvalues are the solutions of

$$0 = (-1/3 - \lambda)^3 - (1/3)(0 - (-1/3 - \lambda)(1/3))$$

Solving for λ

$$0 = (-1/3 - \lambda)[(-1/3 - \lambda)^2 - 1/9]$$

$$0 = (\lambda + 1/3)(-1/3 - \lambda + 1/3)(-1/3 - \lambda - 1/3)$$

$$0 = \lambda(\lambda + 1/3)(\lambda + 2/3)$$

Since all the solutions are less than or equal to 0, the Slutsky matrix is negative semidefinite at $p = (1, 1, 1)$ and $w = 1$.

Question 2.F.10.b. Show that this demand function does not satisfy the weak axiom. [*Hint*: Consider the price vector $(1, 1, \epsilon)$ and show that the substitution matrix is not negative semidefinite (for $\epsilon > 0$ small).]

Answer 2.F.10.b. At $p = (1, 1, \epsilon)$ and $w = 1$, the Slutsky matrix is

$$\begin{bmatrix} -\frac{3+\epsilon}{(2+\epsilon)^2} + \frac{1}{(2+\epsilon)^2} & \frac{1+\epsilon}{(2+\epsilon)^2} + \frac{1}{(2+\epsilon)^2} & -\frac{1}{(2+\epsilon)^2} + \frac{1}{(2+\epsilon)^2} \\ -\frac{1}{(2+\epsilon)^2} + \frac{1}{(2+\epsilon)^2} & -\frac{3+\epsilon}{(2+\epsilon)^2} + \frac{1}{(2+\epsilon)^2} & \frac{1+\epsilon}{(2+\epsilon)^2} + \frac{1}{(2+\epsilon)^2} \\ \frac{1+\epsilon}{\epsilon(2+\epsilon)^2} + \frac{1}{\epsilon(2+\epsilon)^2} & -\frac{1}{\epsilon(2+\epsilon)^2} + \frac{1}{\epsilon(2+\epsilon)^2} & -\frac{2+2\epsilon}{[\epsilon(2+\epsilon)]^2} + \frac{1}{\epsilon(2+\epsilon)^2} \end{bmatrix}$$

Cleaning up this mess, the Slutsky matrix is

$$\begin{bmatrix} -\frac{1}{2+\epsilon} & \frac{1}{2+\epsilon} & 0 \\ 0 & -\frac{1}{2+\epsilon} & \frac{1}{2+\epsilon} \\ \frac{1}{\epsilon(2+\epsilon)} & 0 & -\frac{1}{\epsilon^2(2+\epsilon)} \end{bmatrix}$$

We want to show that for some ϵ this matrix has negative determinant, i.e. two eigenvalues of different signs. Factoring out $\frac{1}{\epsilon^2(2+\epsilon)}$ to make it a little easier, we want to show that for some ϵ

$$\begin{bmatrix} -\epsilon^2 & \epsilon^2 & 0 \\ 0 & -\epsilon^2 & \epsilon^2 \\ \epsilon & 0 & -1 \end{bmatrix}$$

has negative determinant. The determinant is

$$-\epsilon^2(-\epsilon^2)(-1) - \epsilon^2(0 - \epsilon^2\epsilon) = -\epsilon^4 + \epsilon^5$$

Dividing by ϵ^4 , the determinant is negative when $\epsilon < 1$.

Question 2.F.11. Show that for $L = 2$, $S(p, w)$ is always symmetric. [*Hint:* Use Proposition 2.F.3.]

Answer 2.F.11. By Proposition 2.F.3

$$p \cdot S(p, w) = S(p, w)p = 0$$

Expanding

$$\begin{aligned} p_1 s_{11} + p_2 s_{21} + p_1 s_{12} + p_2 s_{22} &= p_1 s_{11} + p_2 s_{12} + p_1 s_{21} + p_2 s_{22} \\ p_2 s_{21} + p_1 s_{12} &= p_2 s_{12} + p_1 s_{21} \\ (p_1 - p_2) s_{12} &= (p_1 - p_2) s_{21} \\ s_{12} &= s_{21} \end{aligned}$$

Since the off diagonal entries are equal, $S(p, w)$ is symmetric.

Question 2.F.12. Show that if the Walrasian demand function $x(p, w)$ is generated by a rational preference relation, then it must satisfy the weak axiom.

Answer 2.F.12. Suppose that $\succeq$ is a rational preference relation generating the Walrasian demand function $x(\cdot)$, and that (p, w) , (p', w') are any two price-wealth combinations, and $p' \cdot x(p, w) \leq w'$. Suppose also that that weak axiom is violated, i.e. $p \cdot x(p', w') \leq w$. Then $x(p, w)$ is revealed preferred to $x(p', w')$ and $x(p', w')$ is revealed preferred to $x(p, w)$. Since $\succeq$ is transitive, $x(p, w) \succeq x(p', w') \succeq x(p, w)$, i.e. $x(p, w) \succ x(p, w)$, contradicting the irreflexivity of $\succeq$ that is an immediate consequence of rationality (Proposition 1.B.1). Therefore, the weak axiom must hold.

Question 2.F.14. Show that if $x(p, w)$ is a Walrasian demand function that satisfies the weak axiom, then $x(p, w)$ must be homogeneous of degree 0.

Answer 2.F.14. By Walras' law, $\alpha p \cdot x(\alpha p, \alpha w) \leq \alpha w$, so that $p \cdot x(\alpha p, \alpha w) \leq w$. By the weak axiom, if $x(p, w) \neq x(\alpha p, \alpha w)$, then $\alpha p \cdot x(p, w) > \alpha w$, so that $p \cdot x(p, w) > w$,

which is a violation of Walras' law. Therefore, $x(p, w) = x(\alpha p, \alpha w)$, i.e. $x(p, w)$ is homogeneous of degree 0.

Question 2.F.16. Consider a setting where $L = 3$ and a consumer whose consumption set is $\mathbb{R}^3$. Suppose that his demand function $x(p, w)$ is

$$\begin{aligned}x_1(p, w) &= \frac{p_2}{p_3} \\x_2(p, w) &= -\frac{p_1}{p_3} \\x_3(p, w) &= \frac{w}{p_3}\end{aligned}$$

Question 2.F.16.a. Show that $x(p, w)$ is homogenous of degree zero in (p, w) and satisfies Walras' law.

Answer 2.F.16.a. Homogeneity of degree zero:

$$\begin{aligned}x_1(\alpha p, \alpha w) &= \frac{\alpha p_2}{\alpha p_3} = \frac{p_2}{p_3} \\x_2(\alpha p, \alpha w) &= -\frac{\alpha p_1}{\alpha p_3} = -\frac{p_1}{p_3} \\x_3(\alpha p, \alpha w) &= \frac{\alpha w}{\alpha p_3} = \frac{w}{p_3}\end{aligned}$$

Walras' Law:

$$p_1 \frac{p_2}{p_3} - p_2 \frac{p_1}{p_3} + p_3 \frac{w}{p_3} = w$$

Question 2.F.16.b. Show that $x(p, w)$ violates the weak axiom.

Answer 2.F.16.b. At $(p, w) = (1, 1, 1, 1)$, $(x_1, x_2, x_3) = (1, -1, 1)$. At $(p, w) = (1, 1, 2, 2)$, $(x_1, x_2, x_3) = (1/2, -1/2, 1)$. Both of these bundles are affordable at other set of prices, so the weak axiom fails.

Question 2.F.16.c. Show that $v \cdot S(p, w)v = 0$ for all $v \in \mathbb{R}^3$.

Answer 2.F.16.c. Computing the Slutsky matrix

$$\begin{aligned}
 \frac{\partial x_1(p, w)}{\partial p_1} &= 0 \\
 \frac{\partial x_1(p, w)}{\partial p_2} &= \frac{1}{p_3} \\
 \frac{\partial x_1(p, w)}{\partial p_3} &= -\frac{p_2}{p_3^2} \\
 \frac{\partial x_1(p, w)}{\partial w} &= 0 \\
 \frac{\partial x_2(p, w)}{\partial p_1} &= -\frac{1}{p_3} \\
 \frac{\partial x_2(p, w)}{\partial p_2} &= 0 \\
 \frac{\partial x_2(p, w)}{\partial p_3} &= \frac{p_1}{p_3^2} \\
 \frac{\partial x_2(p, w)}{\partial w} &= 0 \\
 \frac{\partial x_3(p, w)}{\partial p_1} &= 0 \\
 \frac{\partial x_3(p, w)}{\partial p_2} &= 0 \\
 \frac{\partial x_3(p, w)}{\partial p_3} &= -\frac{w}{p_3^2} \\
 \frac{\partial x_3(p, w)}{\partial w} &= \frac{1}{p_3}
 \end{aligned}$$

Therefore

$$S(p, w) = \begin{bmatrix} 0 & 1/p_3 & -p_2/p_3^2 \\ -1/p_3 & 0 & p_1/p_3^2 \\ 0 & 0 & 0 \end{bmatrix}$$

Now, we show that $v \cdot S(p, w)v = 0$ for any $v \in \mathbb{R}^3$.

$$\begin{aligned}
 \begin{bmatrix} a & b & c \end{bmatrix} \begin{bmatrix} 0 & 1/p_3 & -p_2/p_3^2 \\ -1/p_3 & 0 & p_1/p_3^2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} &= \begin{bmatrix} -b/p_3 & a/p_3 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} \\
 &= 0
 \end{aligned}$$

Question 2.F.17. In an L -commodity world, a consumer's Walrasian demand function is

$$x_k(p, w) = \frac{w}{\sum_{l=1}^L p_l} \text{ for } k = 1, \dots, L$$

Question 2.F.17.a. Is this demand function homogeneous of degree zero in (p, w) ?

Answer 2.F.17.a. Yes, since for any $\alpha > 0$ and any k , $x_k(\alpha p, \alpha w) = \frac{\alpha w}{\sum_{l=1}^L \alpha p_l} = \frac{w}{\sum_{l=1}^L p_l} = x_k(p, w)$.

Question 2.F.17.b. Does it satisfy Walras' law?

Answer 2.F.17.b. Yes, since $\sum_{k=1}^L p_k x_k(p, w) = \frac{w \sum_{k=1}^L p_k}{\sum_{l=1}^L p_l} = w$.

Question 2.F.17.d. Compute the Slutsky substitution matrix for this demand function. Is it negative semidefinite? Negative definite? Symmetric?

Answer 2.F.17.d. The lk th entry of the Slutsky matrix is

$$\begin{aligned} s_{lk}(p, w) &= \frac{\partial x_l(p, w)}{\partial p_k} + \frac{\partial x_l(p, w)}{\partial w} x_k(p, w) \\ &= -\frac{w}{\left(\sum_{l=1}^L p_l\right)^2} + \frac{1}{\sum_{l=1}^L p_l} \frac{w}{\sum_{l=1}^L p_l} \\ &= 0 \end{aligned}$$

So the Slutsky matrix for this demand function is the $L \times L$ matrix of zeroes, which is negative semidefinite and symmetric.

Chapter 3

Question 3.B.1.a. Show that if $\succeq$ is strongly monotone, then it is monotone.

Answer 3.B.1.a. Let x, y be any elements of $X \subseteq \mathbb{R}_+^L$ (a choice set) such that $y \gg x$. Then $y \geq x$ and $y \neq x$. Since $\succeq$ is strongly monotone, $y \succ x$, i.e. $\succeq$ is also monotone.

Question 3.B.1.b. Show that if $\succeq$ is monotone, then it is locally nonsatiated.

Answer 3.B.1.b. Let x be any element of $X \subseteq \mathbb{R}_+^L$ and let $\epsilon > 0$. Let $y = x + \frac{\epsilon}{2\sqrt{L}} i_L$ where i_L is the $L \times 1$ vector of 1s. Then $y \gg x$, so $y \succ x$ since $\succeq$ is monotone, and by construction $\|y - x\| = \sqrt{\left(\frac{\epsilon}{2\sqrt{L}}\right)^2 L} = \epsilon/2 < \epsilon$, so $\succeq$ is locally nonsatiated.

Question 3.B.2. The preference relation $\succeq$ defined on the consumption set $X = \mathbb{R}_+^L$ is said to be *weakly monotone* if and only if $x \geq y$ implies that $x \succeq y$. Show that if $\succeq$ is transitive, locally nonsatiated, and weakly monotone, then it is monotone.

Answer 3.B.2. Let x, y be any elements of X such that $y \gg x$. Let $\epsilon = \min_{i=1, \dots, L} \frac{|y_i - x_i|}{2} < \sqrt{\sum_{i=1}^L |y_i - x_i|^2}$. Since $\succeq$ is locally nonsatiated, there exists $z \in B_\epsilon(x)$ such that $z \succ x$. Since $|z_i - x_i| < |y_i - x_i|$ for all i by construction, $y \geq z$, so $y \succeq z \succ x$. By transitivity, $y \succ x$, i.e. $y \gg x$ implies $y \succ x$, i.e. $\succeq$ is monotone.

Question 3.B.3. Draw a convex preference relation that is locally nonsatiated but is not monotone.

Answer 3.B.3. Preferences given by $u(x, y) = \min\{x, y\}$ have this property.

Question 3.C.1. Verify that the lexicographic ordering is complete, transitive, strongly monotone, and strictly convex.

Answer 3.C.1. Complete since for all any elements $(x_1, x_2), (y_1, y_2) \in \mathbb{R}_+^2$, at least one of the following is true:

- $x_1 > y_1$
- $x_1 < y_1$
- $x_1 = y_1$ and $x_2 > y_2$
- $x_1 = y_1$ and $y_2 > x_2$
- $x_1 = y_1$ and $x_2 = y_2$

Transitive since if $(x_1, x_2) \succeq (y_1, y_2)$ and $(y_1, y_2) \succeq (z_1, z_2)$ then

- $x_1 > y_1$ or $[x_1 = y_1 \text{ and } x_2 \geq y_2]$
- $y_1 > z_1$ or $[y_1 = z_1 \text{ and } y_2 \geq z_2]$

Case 1: $x_1 > y_1$ and $y_1 > z_1$. Then $x_1 > z_1$, so $(x_1, x_2) \succeq (z_1, z_2)$.

Case 2: $x_1 = y_1$, $x_2 \geq y_2$ and $y_1 > z_1$. Then $x_1 > z_1$, so $(x_1, x_2) \succeq (z_1, z_2)$.

Case 3: $x_1 > y_1$, $y_1 = z_1$ and $y_2 \geq z_2$. Then $x_1 > z_1$, so $(x_1, x_2) \succeq (z_1, z_2)$.

Case 4: $x_1 = y_1$, $x_2 \geq y_2$, $y_1 = z_1$ and $y_2 \geq z_2$. Then $x_1 = z_1$ and $x_2 \geq z_2$, so $(x_1, x_2) \succeq (z_1, z_2)$.

Strongly monotone since if $(x_1, x_2) \geq (y_1, y_2)$ and $(x_1, x_2) \neq (y_1, y_2)$.

Case 1: $x_1 > y_1$ and $x_2 \geq y_2$. Then $(x_1, x_2) \succ (y_1, y_2)$.

Case 2: $x_1 = y_1$ and $x_2 > y_2$. Then $(x_1, x_2) \succ (y_1, y_2)$.

Strictly convex since if $(x_1, x_2) \succeq (y_1, y_2)$, $(z_1, z_2) \succeq (y_1, y_2)$, and $0 < \alpha < 1$. We want to show that $\alpha x_1 + (1 - \alpha)z_1 > y_1$ or $[\alpha x_1 + (1 - \alpha)z_1 = y_1$ and $\alpha x_2 + (1 - \alpha)z_2 > y_2]$.

Case 1: $x_1 > y_1$ and $z_1 > y_1$. Then $\alpha x_1 + (1 - \alpha)z_1 > y_1$.

Case 2: $x_1 = y_1$ and $z_1 > y_1$. Then $\alpha x_1 + (1 - \alpha)z_1 > y_1$.

Case 3: $x_1 > y_1$ and $z_1 = y_1$. Then $\alpha x_1 + (1 - \alpha)z_1 > y_1$.

Case 4: $x_1 = y_1 = z_1$ and $x_2 > y_2$ and $z_2 > y_2$. Then $\alpha x_1 + (1 - \alpha)z_1 = y_1$ and $\alpha x_2 + (1 - \alpha)z_2 > y_2$.

In all cases, $\alpha(x_1, x_2) + (1 - \alpha)(z_1, z_2) \succ (y_1, y_2)$, i.e. $\succeq$ is strictly convex.

Question 3.C.2. Show that if u is a continuous utility function representing $\succeq$, then $\succeq$ is continuous.

Answer 3.C.2. Let $\{x^n\}_{n=1}^\infty$ and $\{y^n\}_{n=1}^\infty$ be sequences converging to x and y respectively, and suppose $x^n \succeq y^n$ for all n . Suppose u is continuous utility function representing $\succeq$. Then $u(x^n) \geq u(y^n)$ for all n . Since u is continuous, $\{u(x^n)\}_{n=1}^\infty$ and $\{u(y^n)\}_{n=1}^\infty$ converge to $u(x)$ and $u(y)$ respectively. In order to show that $\succeq$ is continuous, we need to show that $x \succeq y$, i.e. $u(x) \geq u(y)$.

Now, suppose $u(x) < u(y)$. Let $\epsilon = |u(y) - u(x)|/3$. Then there exists $N \in \mathbb{Z}_{>0}$ such that for all $n > N$

- $|u(x^n) - u(x)| < \epsilon$
- $|u(y) - u(y^n)| < \epsilon$

However, then by the triangle inequality

$$|u(x^n) - u(x) + u(y) - u(y^n)| < 2\epsilon$$

By the construction of ϵ and the assumption that $u(x) < u(y)$

$$\begin{aligned} |u(x^n) - u(y^n) + 3\epsilon| &< 2\epsilon \\ -2\epsilon &< u(x^n) - u(y^n) + 3\epsilon < 2\epsilon \\ -5\epsilon &< u(x^n) - u(y^n) < -\epsilon \\ u(x^n) - u(y^n) &< 0 \\ u(x^n) &< u(y^n) \end{aligned}$$

Since u represents $\succeq$, this last inequality implies $y^n \succ x^n$ for $n > N$, contradicting $x^n \succeq y^n$ for all n . Therefore, our assumption that $u(x) < u(y)$ was wrong, i.e. $u(x) \geq u(y)$, i.e. $x \succeq y$. Therefore, $\succeq$ is continuous.

Question 3.C.3. Show that if for every x the upper and lower contour sets $\{y \in \mathbb{R}_+^L | y \succeq x\}$ and $\{y \in \mathbb{R}_+^L | x \succeq y\}$ are closed, then $\succeq$ is continuous according to Definition 3.C.1.

Answer 3.C.3. (From Sinem Kilic): For any x , choose convergent sequences $\{y^n\}_{n=0}^\infty$ converging to y and $\{z^n\}_{n=0}^\infty$ converging to z such that $y_n \in \{w | w \succeq x\}$ for all n and $z_n \in \{w | x \succeq w\}$ for all n . Then $y^n \succeq z^n$ for all n . Further, since closed sets contain all of their limit points, $y \in \{w | w \succeq x\}$ and $z \in \{w | x \succeq w\}$, i.e. $y \succeq z$.

Question 3.C.4. Exhibit an example of a preference relation that is not continuous but is representable by a utility function.

Answer 3.C.4. Define $\succeq$ on $\mathbb{R}_+$ by $5 \succeq x$ for all $x \in \mathbb{R}_+$ and $x \sim y$ for all $x, y \in \mathbb{R}_+ \setminus \{5\}$. Then $\succeq$ can be represented by $u(5) = 1$ and $u(x) = 0$ for all $x \neq 5$.

Question 3.C.5. Establish the following two results.

Question 3.C.5.a. A continuous $\succeq$ is homothetic if and only if it admits a utility function $u(x)$ that is homogeneous of degree one, i.e., $u(\alpha x) = \alpha u(x)$ for all $\alpha > 0$.

Answer 3.C.5.a. Suppose first that $u(x)$ is homogeneous of degree 1 and represents $\succeq$. If $x, y \in \mathbb{R}_+^L$ and $x \sim y$, then $u(x) = u(y)$ and as such for any $\alpha > 0$, $\alpha u(x) = \alpha u(y)$. Since u is homogeneous of degree 1, $u(\alpha x) = u(\alpha y)$, i.e. $\alpha x \sim \alpha y$. So, $x \sim y$ implies $\alpha x \sim \alpha y$ when $\succeq$ admits a utility function $u(x)$ that is homogeneous of degree 1, i.e. $\succeq$ is homothetic.

Now, suppose that $\succeq$ is homothetic. We will construct a representative utility function u that is homogeneous of degree 1. Fix $\bar{x} \in \mathbb{R}_+^L$ and define $u(\bar{x}) = 1$. Now, for any $x \in \mathbb{R}_+^L$, define $u(x) = 1$ if $x \sim \bar{x}$ and $u(x) = \alpha$ if $x \sim \alpha \bar{x}$. Now, we just need to verify three things: (i) u is well-defined, i.e. for any $x \in \mathbb{R}_+^L$, $u(x)$ is unique (ii) u is defined on all of $\mathbb{R}_+^L$ and (iii) u actually represents $\succeq$.

Proof of (i): Suppose that u is not well-defined, i.e. there is some $x \in \mathbb{R}_+^L$ such that $u(x) = \alpha$, $u(x) = \beta$, and $\alpha \neq \beta$, with $\alpha > 0$, $\beta > 0$. So, then $x \sim \alpha \bar{x}$ and $x \sim \beta \bar{x}$. Since $\sim$ is transitive, $\bar{x} \sim \beta \bar{x} / \alpha$. Now, since $\succeq$ is homothetic, it is also monotonic. Therefore, if $\beta / \alpha > 1$, $\beta \bar{x} / \alpha > \bar{x}$, so $\beta \bar{x} / \alpha \succ \bar{x}$; if $\beta / \alpha < 1$, $\bar{x} > \beta \bar{x} / \alpha$, so $\bar{x} \succ \beta \bar{x} / \alpha$. Both of these contradict the earlier conclusion that $\bar{x} \sim \beta \bar{x} / \alpha$, so in fact we have $\alpha = \beta$, so u is well-defined.

Proof of (ii): Suppose that u is not defined on all of $\mathbb{R}_+^L$. Then there exists $x \in \mathbb{R}_+^L$ such that $x \not\sim \alpha \bar{x}$ for any $\alpha > 0$. Consider the ray R through the origin that contains x . Then no point in R is $\sim \bar{x}$ since if there were such a point $y \in R$ with $y \sim x$, we

would have $\alpha y = x$ for some $\alpha > 0$ and therefore $u(x) = \alpha$, i.e. u defined at x . Now, by monotonicity and completeness of $\succeq$, there exists y on the ray such that for all other z on the ray, $z \geq y$, implies $y \succ x$ and for $z \leq y$, implies $x \succ y$. Now, choose sequences $\{y_l^m\}_{m=0}^\infty$ and $\{y_h^m\}_{m=0}^\infty$ so that

- $\{y_l^m\}_{m=0}^\infty \rightarrow z$ and $\{y_h^m\}_{m=0}^\infty \rightarrow z$
- $y_l^m \leq z$ for all m , so $x \succ y_l^m$ for all m
- $y_h^m \geq z$ for all m , so $y_h^m \succ x$ for all m

Then, by continuity, $x \succ z$ and $z \succ x$, which is obviously impossible. Therefore, there is no such x at which u is undefined.

Proof of (iii): Suppose $x \succeq y$ but $\alpha = u(x) < u(y) = \beta$. Then, $x \sim \alpha \bar{x}$ and $y \sim \beta \bar{x}$. But then, by homotheticity, $x \sim \alpha y / \beta$; by monotonicity (remember that $\alpha / \beta < 1$), $y \succ \alpha y / \beta \sim x$; by transitivity, $y \succ x$, a contradiction. So, $x \succeq y$ implies $u(x) \geq u(y)$.

Conversely, suppose $\alpha = u(x) \geq u(y) = \beta$ but $y \succ x$. Then by construction of u , $x \sim \alpha \bar{x}$ and $y \sim \beta \bar{x}$. By homotheticity, $x \sim \alpha y / \beta$; by monotonicity (remember that $\alpha / \beta \geq 1$), $x \sim \alpha y / \beta \succeq y$; by transitivity, $x \succeq y$, a contradiction. So, $u(x) \geq u(y)$ implies $x \succeq y$.

Thus, $u(x) \geq u(y)$ if and only if $x \succeq y$, i.e. u represents $\succeq$ and is homogeneous of degree 1, as desired.

Question 3.C.5.b. A continuous $\succeq$ on $(-\infty, \infty) \times \mathbb{R}_+^{L-1}$ is quasilinear with respect to the first commodity if and only if it admits a utility function $u(x)$ of the form $u(x) = x_1 + \phi(x_2, \dots, x_L)$. [*Hint:* The existence of some continuous utility representation is guaranteed by Proposition 3.C.1.]

Answer 3.C.5.b. Denote e_1 the $L \times 1$ vector $(1, 0, \dots, 0)$. Suppose that $\succeq$ admits $u(x_1, \dots, x_L) = x_1 + \phi(x_2, \dots, x_L)$. If $x \in \mathbb{R}_+^L$, then for $\alpha > 0$, $u(x + \alpha e_1) = \alpha + u(x) > u(x)$, i.e. $\alpha e_1 + x \succ x$. If $x, y \in \mathbb{R}_+^L$ so that $u(x) = u(y)$, then $u(\alpha e_1 + x) = u(\alpha e_1 + y)$, i.e. $\alpha e_1 + x \sim \alpha e_1 + y$. So, $\succeq$ is quasilinear.

Now, suppose that $\succeq$ is quasilinear. We will construct a representative utility function $u(x) = x_1 + \phi(x_2, \dots, x_L)$. First, consider $\succeq_*$, the restriction of $\succeq$ to $\{0\} \times \mathbb{R}_+^{L-1}$. Since $\succeq$ is continuous, $\succeq_*$ is also continuous. Therefore, $\succeq_*$ admits a continuous utility representation, which we will call ϕ . Now, define $u(x) = x_1 + \phi(x_2, \dots, x_L)$. We now show that (i) u is well-defined (ii) u is defined on all of $(-\infty, \infty) \times \mathbb{R}_+^{L-1}$ and (iii) u actually represents $\succeq$.

Proof of (i): Suppose that u is not well-defined, i.e. there is some $x \in (-\infty, \infty) \times \mathbb{R}_+^{L-1}$ such that $u(x) = \alpha$, $u(x) = \beta$, and $\alpha \neq \beta$. Then $x \sim \alpha e_1 + (0, x_2, \dots, x_L)$ and $x \sim \beta e_1 + (0, x_2, \dots, x_L)$, i.e. $\alpha = \beta$ after all, i.e. u is well-defined.

Proof of (ii): By construction, u is defined on $(-\infty, \infty) \times \mathbb{R}_+^{L-1}$.

Proof if (iii): Suppose $x \succeq y$ but $\alpha = u(x) < u(y) = \beta$. Then $u((\beta - \alpha)e_1 + x) = \beta = u(y)$, which implies $(\beta - \alpha)e_1 + x \sim y$. However, since $\beta - \alpha > 0$, $y \sim (\beta - \alpha)e_1 + x \succ x$ and by transitivity $y \succ x$, a contradiction. So, $x \succeq y$ implies $u(x) \geq u(y)$.

Conversely, suppose $\alpha = u(x) \geq u(y) = \beta$ but $y \succ x$. Then by construction of $u((\alpha - \beta)e_1 + y) = \alpha = u(x)$, which implies $(\alpha - \beta)e_1 + y \sim x$. However, since $\alpha - \beta \geq 0$, $x \sim (\alpha - \beta)e_1 + y \succeq y$ and by transitivity $x \succeq y$, a contradiction. So, $u(x) \geq u(y)$ implies $x \succeq y$.

Thus, $u(x) \geq u(y)$ if and only if $x \succeq y$, i.e. u represents $\succeq$ and $u(x) = x_1 + \phi(x_2, \dots, x_L)$.

Question 3.C.6. Suppose that in a two commodity world, the consumer's utility function takes the form $u(x) = [\alpha_1 x_1^\rho + \alpha_2 x_2^\rho]^{1/\rho}$. This utility function is known as the *constant elasticity of substitution* (or *CES*) utility function.

Question 3.C.6.a. Show that when $\rho = 1$, the indifference curves become linear.

Answer 3.C.6.a. When $\rho = 1$, $u(x) = \alpha_1 x_1 + \alpha_2 x_2$ which is linear for any fixed level of $u(x)$.

Question 3.C.6.b. Show that as $\rho \rightarrow 0$, this utility function comes to represent the same preferences as the (generalized) Cobb-Douglas utility function $u(x) = x_1^{\alpha_1} x_2^{\alpha_2}$.

Answer 3.C.6.b. First, let's modify both utility functions by taking monotonic transformations of each, which does not change the preferences. Thus, we can represent CES preferences by $u_{CES}(x) = \frac{\alpha_1 + \alpha_2}{\rho} \ln[\alpha_1 x_1^\rho + \alpha_2 x_2^\rho]$ and Cobb-Douglas preferences by $u_{CD}(x) = \alpha_1 \ln x_1 + \alpha_2 \ln x_2$. Now, note that in the limit, these functions are the same, therefore in the limit they represent the same preferences.

$$\lim_{\rho \rightarrow 0} u_{CES}(x) = \lim_{\rho \rightarrow 0} \frac{\alpha_1 + \alpha_2}{\rho} \ln[\alpha_1 x_1^\rho + \alpha_2 x_2^\rho]$$

By L'Hôpital's Rule

$$\begin{aligned}
 &= \lim_{\rho \rightarrow 0} (\alpha_1 + \alpha_2) \frac{\alpha_1 x_1^\rho \ln x_1 + \alpha_2 x_2^\rho \ln x_2}{\alpha_1 x_1^\rho + \alpha_2 x_2^\rho} \\
 &= \alpha_1 \ln x_1 + \alpha_2 \ln x_2 \\
 &= u_{CD}(x)
 \end{aligned}$$

Question 3.C.6.c. Show that as $\rho \rightarrow -\infty$, indifference curves become “right angles”.

Answer 3.C.6.c. We want to show that for a given utility level, $\lim_{\rho \rightarrow -\infty} \frac{dx_2}{dx_1} = 0$ if $x_1 < x_2$ and $\lim_{\rho \rightarrow -\infty} \frac{dx_2}{dx_1} = \infty$ if $x_2 < x_1$.

First, differentiate $k = [\alpha_1 x_1^\rho + \alpha_2 x_2^\rho]^{1/\rho}$ with respect to x_1 :

$$\begin{aligned}
 k &= [\alpha_1 x_1^\rho + \alpha_2 x_2^\rho]^{1/\rho} \\
 0 &= \frac{1}{\rho} [\alpha_1 x_1^\rho + \alpha_2 x_2^\rho]^{\frac{1-\rho}{\rho}} \left[\alpha_1 \rho x_1^{\rho-1} + \alpha_2 \rho x_2^{\rho-1} \frac{dx_2}{dx_1} \right]
 \end{aligned}$$

Since $[\alpha_1 x_1^\rho + \alpha_2 x_2^\rho]^{\frac{1-\rho}{\rho}} = 0$ only if $x_1 = x_2 = 0$:

$$\begin{aligned}
 0 &= \alpha_1 x_1^{\rho-1} + \alpha_2 x_2^{\rho-1} \frac{dx_2}{dx_1} \\
 \frac{dx_2}{dx_1} &= -\frac{\alpha_1 x_1^{\rho-1}}{\alpha_2 x_2^{\rho-1}}
 \end{aligned}$$

Now, find $\lim_{\rho \rightarrow -\infty} \frac{dx_2}{dx_1} = 0$.

$$\begin{aligned}
 \lim_{\rho \rightarrow -\infty} \frac{dx_2}{dx_1} &= \lim_{\rho \rightarrow -\infty} -\frac{\alpha_1 x_1^{\rho-1}}{\alpha_2 x_2^{\rho-1}} \\
 &= \lim_{\rho \rightarrow -\infty} -\frac{\alpha_1}{\alpha_2} \left(\frac{x_1}{x_2} \right)^{\rho-1} \\
 &= 0 \text{ if } x_1 < x_2 \\
 &= \infty \text{ if } x_2 < x_1
 \end{aligned}$$

Question 3.D.1. Verify the three properties of Proposition 3.D.2 for the Walrasian demand function generated by the Cobb-Douglas utility function.

Answer 3.D.1. Recall that for (p, w) and $u(x) = \prod_{i=1}^L x_i^{\alpha_i}$, Walrasian demand is $x(p, w) = (\alpha_1 w/p_1, \dots, \alpha_L w/p_L)$ (where, without loss of generality, we assume that $\sum_{i=1}^L \alpha_i = 1$). Then

Walrasian demand is homogeneous of degree 0 since for $\beta > 0$,

$$\begin{aligned} x(\beta p, \beta w) &= (\beta \alpha_1 w / (\beta p_1), \dots, \beta \alpha_L w / (\beta p_L)) \\ &= (\alpha_1 w / p_1, \dots, \alpha_L w / p_L) \\ &= x(p, w) \end{aligned}$$

Walrasian demand satisfies Walras' law since $p \cdot x(p, w) = \sum_{i=1}^L p_i \alpha_i w / p_i = w \sum_{i=1}^L \alpha_i = w$.

Walrasian demand is unique for any (p, w) by inspection.

Question 3.D.2. Verify the four properties of Proposition 3.D.3 for the indirect utility function derived in Example 3.D.2.

Answer 3.D.2. The indirect utility function derived in Example 3.D.2 is for Cobb-Douglas utility in two goods:

$$\begin{aligned} v(p, w) &= [\alpha \ln \alpha + (1 - \alpha) \ln(1 - \alpha)] + \ln w - \alpha \ln p_1 - (1 - \alpha) \ln p_2 \\ &= \alpha \ln \frac{\alpha w}{p_1} + (1 - \alpha) \ln \frac{(1 - \alpha) w}{p_2} \end{aligned}$$

The indirect utility function is homogeneous of degree 0 since for $\beta > 0$, $v(\beta p, \beta w) = \alpha \ln \frac{\beta \alpha w}{\beta p_1} + (1 - \alpha) \ln \frac{\beta(1 - \alpha) w}{\beta p_2} = v(p, w)$.

The indirect utility function is strictly increasing in w since $\frac{\partial v(p, w)}{\partial w} = 1/w > 0$ for $w > 0$.

The indirect utility function is nonincreasing in p_1 since $\frac{\partial v(p, w)}{\partial p_1} = -\alpha/p_1 < 0$ for $p_1 > 0$.

The indirect utility function is nonincreasing in p_2 since $\frac{\partial v(p, w)}{\partial p_2} = -(1 - \alpha)/p_2 < 0$ for $p_2 > 0$.

The indirect utility function is quasiconvex. For any constant $\bar{v}$ and any two (p, w) and (p', w') such that $v(p, w) \leq \bar{v}$ and $v(p', w') \leq \bar{v}$, for any $0 \leq \beta \leq 1$, put $(p'', w'') = (\beta p + (1 - \beta)p', \beta w + (1 - \beta)w')$. We want to show that $v(p'', w'') \leq \bar{v}$. Since

$$v(p'', w'') = [\alpha \ln \alpha + (1 - \alpha) \ln(1 - \alpha)] + \ln w'' - \alpha \ln p_1'' - (1 - \alpha) \ln p_2''$$

Since $\ln$ is concave

$$\begin{aligned} &\leq [\alpha \ln \alpha + (1 - \alpha) \ln(1 - \alpha)] + \beta \ln w + (1 - \beta) \ln w' \\ &\quad - \alpha(\beta \ln p_1 + (1 - \beta) \ln p'_1) - (1 - \alpha)(\beta \ln p_2 + (1 - \beta) \ln p'_2) \\ &\leq \beta v(p, w) + (1 - \beta)v(p', w') \end{aligned}$$

The indirect utility function is continuous in (p, w) since it is the sum of continuous functions in (p, w) .

Question 3.D.3. Suppose that $u(x)$ is differentiable and strictly quasiconcave and that the Walrasian demand function $x(p, w)$ is differentiable. Show the following:

Question 3.D.3.a. If $u(x)$ is homogeneous of degree 1, then the Walrasian demand function $x(p, w)$ and the indirect utility function $v(p, w)$ are homogeneous of degree 1 [and hence can be written in the form $x(p, w) = w\tilde{x}(p)$ and $v(p, w) = w\tilde{v}(p)$] and the wealth expansion path (see Section 2.E) is a straight line through the origin. What does this imply about the wealth elasticities of demand?

Answer 3.D.3.a. See also Exercise 2.E.4. First, show that $x(p, w)$ is homogeneous of degree 1 in w . Suppose $x(p, w) = (\bar{x}_1, \dots, \bar{x}_L)$. Then since u is differentiable, $(\bar{x}_1, \dots, \bar{x}_L)$ solves the following system of equations (the first order conditions and constraint of the utility maximization problem for (p, w)):

$$\begin{aligned} 0 &= \frac{\partial u}{\partial x_l} \Big|_{(\bar{x}_1, \dots, \bar{x}_L)} - \lambda p_l, \quad l = 1, \dots, L \\ 0 &= w - p \cdot (\bar{x}_1, \dots, \bar{x}_L) \end{aligned}$$

Since $u(x)$ is homogeneous of degree 1, $\frac{\partial u}{\partial x_l}$ is homogeneous of degree 0 for all l . So, for any $\alpha > 0$

$$\begin{aligned} 0 &= \frac{\partial u}{\partial x_l} \Big|_{(\alpha \bar{x}_1, \dots, \alpha \bar{x}_L)} - \lambda p_l, \quad l = 1, \dots, L \\ 0 &= \alpha w - p \cdot (\alpha \bar{x}_1, \dots, \alpha \bar{x}_L) \end{aligned}$$

However, these conditions are simply the first order conditions and constraint from the utility maximization problem for $(p, \alpha w)$, i.e. $\alpha x(p, w)$ solves the $(p, \alpha w)$ problem. Therefore, $x(p, \alpha w) = \alpha x(p, w)$, i.e. $x(p, w)$ is homogeneous of degree 1 in w .

Now, show that $v(p, w)$ is homogeneous of degree 1 in w since by homogeneity of degree 1 of $x(p, w)$ and $u(x)$, $v(p, \alpha w) = u(x(p, \alpha w)) = u(\alpha x(p, w)) = \alpha u(x(p, w)) = \alpha v(p, w)$.

Refer to the result from Exercise 2.E.4 to conclude that the wealth expansion path is a straight line through the origin, and that the wealth elasticities of demand are therefore 1 for every good.

Question 3.D.3.b. If $u(x)$ is strictly quasiconcave and $v(p, w)$ is homogeneous of degree one in w , then $u(x)$ must be homogeneous of degree one.

Answer 3.D.3.b. We're going to cheat here and use Proposition 3.G.4 (Roy's identity). Note that at this point in the text we have all the tools necessary to complete Proof 2 of Roy's identity from Mas-Colell, so it's not that big a cheat. A quick check of the official solutions revealed that they too take this shortcut.

$$x(p, \alpha w) = -\frac{\nabla_p v(p, \alpha w)}{\nabla_w v(p, \alpha w)}$$

Since $v(p, w)$ is homogeneous of degree 1 in w , $\nabla_p v(p, w)$ is homogeneous of degree 1 in w , and $\nabla_w v(p, w)$ is homogeneous of degree 0 in w . Thus

$$\begin{aligned} &= -\frac{\alpha \nabla_p v(p, w)}{\nabla_w v(p, w)} \\ &= \alpha x(p, w) \end{aligned}$$

In other words, $x(p, w)$ is homogeneous of degree 1 in w . Then

$$\begin{aligned} \alpha u(x(p, w)) &= \alpha v(p, w) \\ &= v(p, \alpha w) \\ &= u(x(p, \alpha w)) \\ &= u(\alpha x(p, w)) \end{aligned}$$

Question 3.D.4. Let $(-\infty, \infty) \times \mathbb{R}_+^{L-1}$ denote the consumption set, and assume that preferences are strictly convex and quasilinear. Normalize $p_1 = 1$.

Question 3.D.4.a. Show that the Walrasian demand functions for goods $2, \dots, L$ are independent of wealth. What does this imply about the wealth effect (see Section 2.E) of demand for good 1?

Answer 3.D.4.a. By Exercise 3.C.5, there exists a continuous function $\bar{\phi}$ such that $u(x) = x_1 + \bar{\phi}(x_2, \dots, x_L)$ represents the preferences. Then the first order conditions of the UMP are

$$\begin{aligned} 0 &= 1 - \lambda \text{ (for good 1)} \\ 0 &= \frac{\partial \bar{\phi}}{\partial x_l} - \lambda p_l \text{ (for goods } 2, \dots, L) \end{aligned}$$

Substituting $\lambda = 1$ into the latter:

$$0 = \frac{\partial \bar{\phi}}{\partial x_l} - p_l \text{ (for goods } 2, \dots, L)$$

These $L-1$ equations are independent of w and in $L-1$ unknowns, namely $x_2, \dots, x_L$, so that demand for goods $2, \dots, L$ is independent of w .

Question 3.D.4.b. Argue that the indirect utility function can be written in the form $v(p, w) = w + \phi(p)$ for some function $\phi(\cdot)$.

Answer 3.D.4.b. Observe from the conditions above and the budget constraint that

$$x_1(p, w) = w - \sum_{l=2}^L \frac{\partial \bar{\phi}}{\partial x_l} x_l$$

Substituting into the utility function

$$\begin{aligned} v(p, w) &= u(x(p, w)) \\ &= w - \underbrace{\sum_{l=2}^L \frac{\partial \bar{\phi}}{\partial x_l} x_l + \bar{\phi}(x_2(p), \dots, x_L(p))}_{\phi(p)} \\ &= w + \phi(p) \end{aligned}$$

Here are the “official” solutions, cleaned up, but without the assumption that u is differentiable.

Question 3.D.4.c. Suppose, for simplicity, that $L = 2$, and write the consumer’s utility function as $u(x_1, x_2) = x_1 + \eta(x_2)$. Now, however, let the consumption set be $\mathbb{R}_+^2$ so that there is a nonnegativity constraint on consumption of the numeraire x_1 . Fix prices p , and examine how the consumer’s Walrasian demand changes as wealth w varies. When is the nonnegativity constraint on the numeraire irrelevant?

Answer 3.D.4.c. By Exercise 3.D.4.a, $x_2(p, w)$ is independent of wealth, so we can write it $x_2(p)$. Therefore, at fixed prices, the ideal consumption level of good 2 is fixed. If the consumer can consume a negative amount of good 1, she will always consume $x_2(p)$ at prices p even if $p \cdot x_2(p) > w$. However, if she cannot borrow, the nonnegativity constraint on good 1 will bind whenever $p \cdot x_2(p) > w$.

Question 3.D.5. Consider again the CES utility function of Exercise 3.C.6, and assume that $\alpha_1 = \alpha_2 = 1$.

Question 3.D.5.a. Compute the Walrasian demand and indirect utility functions for this utility function.

Answer 3.D.5.a. Transform $\tilde{u}(x) = [x_1^\rho + x_2^\rho]^{1/\rho}$ by $u(x) = [\tilde{u}(x)]^\rho = x_1^\rho + x_2^\rho$. This transformation is monotonic, so it does not change the preferences.

The UMP is

$$\max_{x_1, x_2} x_1^\rho + x_2^\rho \text{ subject to } p_1 x_1 + p_2 x_2 \leq w$$

The first order conditions (with multiplier λ) are

$$\begin{aligned} 0 &= \rho x_1^{\rho-1} - \lambda p_1 \\ 0 &= \rho x_2^{\rho-1} - \lambda p_2 \end{aligned}$$

Eliminating λ

$$x_1 = \left(\frac{p_1}{p_2} \right)^{1/(\rho-1)} x_2$$

Substituting into the budget constraint

$$\begin{aligned} w &= p_1 \left(\frac{p_1}{p_2} \right)^{1/(\rho-1)} x_2 + p_2 x_2 \\ x_2 &= \frac{w p_2^{1/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}} \end{aligned}$$

By symmetry

$$x_1 = \frac{w p_1^{1/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}}$$

In summary,

$$x(p, w) = \left(\frac{w p_2^{1/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}}, \frac{w p_1^{1/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}} \right)$$

The indirect utility function is

$$\begin{aligned} v(p, w) &= \left[\frac{wp_1^{1/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}} \right]^\rho + \left[\frac{wp_2^{1/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}} \right]^\rho \\ &= \frac{w^\rho}{\left(p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)} \right)^{\rho-1}} \end{aligned}$$

Question 3.D.5.b. Verify that these two functions satisfy all the properties of Propositions 3.D.2 and 3.D.3.

Answer 3.D.5.b. Walrasian demand is homogeneous of degree 0.

$$\begin{aligned} x(\alpha p, \alpha w) &= \left(\frac{\alpha w (\alpha p_2)^{1/(\rho-1)}}{(p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)})}, \frac{wp_1^{1/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}} \right) \\ &= \left(\frac{wp_2^{1/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}}, \frac{wp_1^{1/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}} \right) \end{aligned}$$

Walras' law holds.

$$\begin{aligned} p \cdot x(p, w) &= p_1 \frac{wp_1^{1/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}} + p_2 \frac{wp_2^{1/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}} \\ &= \frac{wp_1^{\rho/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}} + \frac{wp_2^{\rho/(\rho-1)}}{p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}} \\ &= w \end{aligned}$$

The utility function is strictly quasiconcave and $x(p, w)$ is a unique element (uniqueness is by inspection of the demand rules). Put $(x_1'', x_2'') = \alpha(x_1, x_2) + (1 - \alpha)(x_1', x_2')$ where (x_1, x_2) and (x_1', x_2') are arbitrary points in $\mathbb{R}_+^L$ and $\alpha \in (0, 1)$. We show that $u(x_1'', x_2'') > \min\{u(x_1, x_2), u(x_1', x_2')\}$.

$$\begin{aligned} u(x_1'', x_2'') &= [\alpha x_1 + (1 - \alpha)x_1']^\rho + [\alpha x_2 + (1 - \alpha)x_2']^\rho \\ &> (\alpha x_1)^\rho + ((1 - \alpha)x_1')^\rho + (\alpha x_2)^\rho + ((1 - \alpha)x_2')^\rho \\ &> \alpha^\rho u(x_1, x_2) + (1 - \alpha)^\rho u(x_1', x_2') \\ &> \min\{u(x_1, x_2), u(x_1', x_2')\} \end{aligned}$$

The indirect utility function is homogeneous of degree zero.

$$\begin{aligned}
 v(\alpha p, \alpha w) &= \frac{(\alpha w)^\rho}{((\alpha p_1)^{\rho/(\rho-1)} + (\alpha p_2)^{\rho/(\rho-1)})^{\rho-1}} \\
 &= \frac{\alpha^\rho w^\rho}{\left[\alpha^{\rho/(\rho-1)} \left(p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)} \right) \right]^{\rho-1}} \\
 &= \frac{w^\rho}{\left(p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)} \right)^{\rho-1}} \\
 &= v(p, w)
 \end{aligned}$$

The indirect utility function is strictly increasing in w .

$$\begin{aligned}
 \frac{\partial v(p, w)}{\partial w} &= \frac{\rho w^{\rho-1}}{\left(p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)} \right)^{\rho-1}} \\
 &> 0
 \end{aligned}$$

The indirect utility function is nonincreasing in p_1, p_2 .

$$\begin{aligned}
 \frac{\partial v(p, w)}{\partial p_1} &= w^\rho (1 - \rho) \left(p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)} \right)^{-\rho} \frac{\rho}{\rho - 1} p_1^{1/(\rho-1)} \\
 &= w^\rho \rho \left(p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)} \right)^{-\rho} p_1^{1/(\rho-1)} \\
 &\geq 0 \\
 \frac{\partial v(p, w)}{\partial p_2} &= w^\rho (1 - \rho) \left(p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)} \right)^{-\rho} \frac{\rho}{\rho - 1} p_2^{1/(\rho-1)} \\
 &= w^\rho \rho \left(p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)} \right)^{-\rho} p_2^{1/(\rho-1)} \\
 &\geq 0
 \end{aligned}$$

The indirect utility function is quasiconvex. Suppose that (p_1, p_2, w) and (p'_1, p'_2, w') are two points such that $v(p, w) \leq \bar{v}$ where $\bar{v}$ is some fixed utility level. Let $\alpha \in (0, 1)$

and $(p_1'', p_2'', w'') = \alpha(p_1, p_2, w) + (1 - \alpha)(p_1', p_2', w')$.

$$\begin{aligned} v(p_1'', p_2'', w'') &= \frac{(\alpha w + (1 - \alpha)w')^\rho}{\{(\alpha p_1 + (1 - \alpha)p_1')^{\rho/(\rho-1)} + (\alpha p_2 + (1 - \alpha)p_2')^{\rho/(\rho-1)}\}^{\rho-1}} \\ &\leq \frac{\min\{w^\rho, w'^\rho\}}{\max\left\{\left(p_1^{\rho/(\rho-1)} + p_2^{\rho/(\rho-1)}\right)^{\rho-1}, \left(p_1'^{\rho/(\rho-1)} + p_2'^{\rho/(\rho-1)}\right)^{\rho-1}\right\}} \\ &\leq \bar{v} \end{aligned}$$

Question 3.D.6. Consider the three-good setting in which the consumer has utility function $u(x) = (x_1 - b_1)^\alpha (x_2 - b_2)^\beta (x_3 - b_3)^\gamma$.

Question 3.D.6.a. What can you assume that $\alpha + \beta + \gamma = 1$ without loss of generality? Do so for the rest of the problem.

Answer 3.D.6.a. We can take a monotonic transformation of u which does not change the preferences by

$$\tilde{u}(x) = [u(x)]^{1/(\alpha+\beta+\gamma)} = (x_1 - b_1)^{\alpha/(\alpha+\beta+\gamma)} (x_2 - b_2)^{\beta/(\alpha+\beta+\gamma)} (x_3 - b_3)^{\gamma/(\alpha+\beta+\gamma)}$$

Question 3.D.6.b. Write down the first-order conditions for the UMP, and derive the consumer's Walrasian demand and indirect utility functions. This system of demands is known as the *linear expenditure system* and is due to Stone (1954).

Answer 3.D.6.b. Without loss of generality, the UMP is

$$\max_{x_1, x_2, x_3} \alpha \ln(x_1 - b_1) + \beta \ln(x_2 - b_2) + \gamma \ln(x_3 - b_3) \text{ s.t. } p_1 x_1 + p_2 x_2 + p_3 x_3 \leq w$$

The Lagrangian is

$$L = \alpha \ln(x_1 - b_1) + \beta \ln(x_2 - b_2) + \gamma \ln(x_3 - b_3) + \lambda(w - p_1 x_1 - p_2 x_2 - p_3 x_3)$$

Utility is increasing in all goods, so the budget constraint binds. The first order conditions are

$$\begin{aligned} 0 &= \frac{\alpha}{x_1 - b_1} - \lambda p_1 \\ 0 &= \frac{\beta}{x_2 - b_2} - \lambda p_2 \\ 0 &= \frac{\gamma}{x_3 - b_3} - \lambda p_3 \\ 0 &= w - p_1 x_1 - p_2 x_2 - p_3 x_3 \end{aligned}$$

Solving this system:

Eliminating λ

$$\begin{aligned} p_1\beta(x_1 - b_1) &= p_2\alpha(x_2 - b_2) \\ p_1\gamma(x_1 - b_1) &= p_3\alpha(x_3 - b_3) \\ 0 &= w - p_1x_1 - p_2x_2 - p_3x_3 \end{aligned}$$

Changing variables by $\tilde{x}_i = x_i - b_i$:

$$\begin{aligned} p_1\beta\tilde{x}_1 &= p_2\alpha\tilde{x}_2 \\ p_1\gamma\tilde{x}_1 &= p_3\alpha\tilde{x}_3 \\ 0 &= w - p_1(\tilde{x}_1 + b_1) - p_2(\tilde{x}_2 + b_2) - p_3(\tilde{x}_3 + b_3) \end{aligned}$$

Eliminating $\tilde{x}_2$ and $\tilde{x}_3$:

$$0 = w - p_1(\tilde{x}_1 + b_1) - p_2\left(\frac{p_1\beta\tilde{x}_1}{p_2\alpha} + b_2\right) - p_3\left(\frac{p_1\gamma\tilde{x}_1}{p_3\alpha} + b_3\right)$$

Solving for $\tilde{x}_1$:

$$\begin{aligned} 0 &= \alpha w - \alpha p_1(\tilde{x}_1 + b_1) - (p_1\beta\tilde{x}_1 + \alpha p_2b_2) - (p_1\gamma\tilde{x}_1 + \alpha p_3b_3) \\ 0 &= \alpha(w - p_1b_1 - p_2b_2 - p_3b_3) - \tilde{x}_1(p_1\alpha + p_1\beta + p_1\gamma) \\ \tilde{x}_1 &= \frac{\alpha(w - p_1b_1 - p_2b_2 - p_3b_3)}{p_1} \end{aligned}$$

Changing variables back:

$$x_1 = \frac{\alpha(w - p_1b_1 - p_2b_2 - p_3b_3)}{p_1} + b_1$$

By symmetry

$$\begin{aligned} x_2 &= \frac{\beta(w - p_1b_1 - p_2b_2 - p_3b_3)}{p_2} + b_2 \\ x_3 &= \frac{\gamma(w - p_1b_1 - p_2b_2 - p_3b_3)}{p_3} + b_3 \end{aligned}$$

Question 3.D.6.c. Verify that these demand functions satisfy the properties listed in Proposition 3.D.2 and 3.D.3.

Answer 3.D.6.c. Homogeneity of degree 0 : It is sufficient to verify this property solely for x_1 due to the symmetry. For $a > 0$, $x_1(ap, aw) = \frac{\alpha(aw - ap_1b_1 - ap_2b_2 - ap_3b_3)}{ap_1} + b_1 = \frac{\alpha(w - p_1b_1 - p_2b_2 - p_3b_3)}{p_1} + b_1 = x_1(p, w)$.

Walras' law: Since $\alpha + \beta + \gamma = 1$, $p \cdot x(p, w) = \sum_{i=1}^3 p_i \left(\frac{\alpha(w - p_1 b_1 - p_2 b_2 - p_3 b_3)}{p_i} + b_i \right) = w$.

Walrasian demand is unique for any (p, w) by inspection.

Since u represents locally nonsatiated preferences and is continuous, all the properties of Proposition 3.D.3. hold.

Question 3.D.8. Show that for all (p, w) , $w \partial v(p, w) / \partial w = -p \nabla_p v(p, w)$.

Answer 3.D.8. Since $v(p, w)$ is homogeneous of degree 0 (Proposition 3.D.3), by Euler's formula

$$\begin{aligned} 0 &= w \frac{\partial v(p, w)}{\partial w} + \sum_{l=1}^L p_l \frac{\partial v(p, w)}{\partial p_l} \\ 0 &= w \frac{\partial v(p, w)}{\partial w} + p \nabla_p v(p, w) \\ w \frac{\partial v(p, w)}{\partial w} &= -p \nabla_p v(p, w) \end{aligned}$$

Question 3.E.1. Assume that $u(\cdot)$ is differentiable. Show that the first-order conditions for the EMP are

$$p \geq \lambda \nabla u(x^*)$$

and

$$x^* \cdot [p - \lambda \nabla u(x^*)] = 0$$

Answer 3.E.1. The EMP is

$$\min_x p \cdot x \text{ subject to } u(x) \geq \bar{u} \text{ and non-negativity constraints on } x_1, \dots, x_L$$

The FOC (with multiplier λ) for x_l is

$$p_l - \lambda \frac{\partial u}{\partial x_l} \geq 0$$

Together with the non-negativity constraints stacking over all goods:

$$\begin{aligned} 0 &= x^* \cdot [p - \lambda \nabla u(x^*)] \\ p &\geq \lambda \nabla u(x^*) \end{aligned}$$

Question 3.E.3. Prove that a solution to the EMP exists if $p \gg 0$ and there is some $x \in \mathbb{R}_+^L$ satisfying $u(x) \geq \bar{u}$.

Answer 3.E.3. It is instructive to compare this claim to the parallel claim and proof for the UMP, quoted here from MWG, p. 50 - 51:

Proposition 3.D.1: If $p \gg 0$ and $u(\cdot)$ is continuous, then the utility maximization problem has a solution.

Proof: If $p \gg 0$, then the budget set $B_{p,w} = \{x \in \mathbb{R}_+^L | p \cdot x \leq w\}$ is a compact set because it is both bounded and closed. The result follows from the fact that a continuous function always has a maximum value on any compact set. (see Section M.F of the Mathematical Appendix).

Now, we can't directly apply this line of reasoning to the UMP, since the set $\{x \in \mathbb{R}_+^L | u(x) \geq u\}$ may not be (in fact, usually will not be) compact. However, we have an extra piece of information for the EMP which has no parallel in the UMP, namely, that there is some $x \in \mathbb{R}_+^L$ satisfying $u(x) \geq u$. The idea of the proof is to use that extra piece of information to further limit the constraint set of the EMP to a compact set, and then follow the same line of reasoning as the proof of Proposition 3.D.1. Here's how to do it.

Let $\tilde{x}$ be any element of $\{x \in \mathbb{R}_+^L | u(x) \geq u\}$. Then, $e(p, u) \leq p \cdot \tilde{x}$. Now, define $B = \{x \in \mathbb{R}_+^L | p \cdot x \leq p \cdot \tilde{x}\}$. It's immediately clear that (i) B is closed and bounded, hence compact and (ii) if the EMP has a solution, it must be an element of B , since all elements of B do at least as well minimizing expenditure as $\tilde{x}$, which also satisfies the constraint $u(x) \geq u$. So, the EMP is equivalent to minimizing expenditure on $B \cap \{x \in \mathbb{R}_+^L | u(x) \geq u\}$. Since $B \cap \{x \in \mathbb{R}_+^L | u(x) \geq u\}$ an intersection of compact sets, $B \cap \{x \in \mathbb{R}_+^L | u(x) \geq u\}$ is compact. Now, u is continuous, so it must achieve a minimum value on a compact set, so the EMP has a solution.

Question 3.E.4. Show that if the consumer's preferences $\succeq$ are convex, then $h(p, u)$ is a convex set. Also, show that if $u(x)$ is strictly convex, then $h(p, u)$ is single-valued.

Answer 3.E.4. Suppose $\succeq$ are convex, and $x = (x_1, \dots, x_L)$, $x' = (x'_1, \dots, x'_L) \in h(p, u)$. We want to show that for any $\alpha \in [0, 1]$, $x'' = \alpha x + (1 - \alpha)x' \in h(p, u)$. Since $x = (x_1, \dots, x_L)$, $x' = (x'_1, \dots, x'_L) \in h(p, u)$, $u(x) \geq u$ and $u(x') \geq u$. Since the preferences are convex, u must be quasiconcave, so $u(x'') \geq \min\{u(x), u(x')\} \geq u$. Finally, since $x = (x_1, \dots, x_L)$, $x' = (x'_1, \dots, x'_L) \in h(p, u)$, for any price level p , $p \cdot x'' = p \cdot \alpha x + p \cdot (1 - \alpha)x' = \alpha e(p, u) + (1 - \alpha)e(p, u) = e(p, u)$. Since $p \cdot x'' = e(p, u)$ and $u(x'') \geq u$, x'' must also solve the EMP, i.e. $x'' \in h(p, u)$.

Suppose further that $\succeq$ are convex. Then the solution to the UMP must be unique at any given (p, w) . Then by duality, $h(p, v(p, w)) = h(p, u)$ must also be unique.

Question 3.E.7. Show that if $\succeq$ is quasilinear with respect to good 1, the Hicksian demand functions for goods $2, \dots, L$ do not depend on u . What is the form of the expenditure function in this case?

Answer 3.E.7. From Exercise 3.E.5, $\succeq$ has a representative utility function $u(x) = x_1 + \phi(x_2, \dots, x_L)$. The first order conditions of the EMP (assuming that the constraint binds) are

$$\begin{aligned} 0 &= p_1 - \lambda \\ 0 &= p_l - \lambda \frac{\partial u}{\partial x_l} \text{ for } l = 2, \dots, L \end{aligned}$$

Substituting $\lambda = p_1$

$$0 = p_l - p_1 \frac{\partial u}{\partial x_l} \text{ for } l = 2, \dots, L$$

which constitute $L - 1$ equations in $L - 1$ unknowns (namely, $x_2, \dots, x_L$) that are independent of the reservation utility level u . (Note that the reservation utility level is different from the function $u(x)$).

Question 3.E.9. Use the relations in (3.E.1) to show that the properties of the indirect utility function identified in Proposition 3.D.3 imply Proposition 3.E.2. Likewise, use the relations in (3.E.1) to prove that Proposition 3.E.2 implies Proposition 3.D.3.

Answer 3.E.9. Part (i): The indirect utility function is homogeneous of degree 0 if and only if the expenditure function is homogeneous of degree 1 in p .

For $\alpha > 0$

$$\begin{aligned} e(\alpha p, u) &= e(\alpha p, v(\alpha p, \alpha w)) \text{ where } u = v(\alpha p, \alpha w) \\ &= \alpha w \text{ by duality} \\ &= \alpha e(p, v(p, w)) \text{ by duality} \\ &= \alpha e(p, u) \text{ by homogeneity of degree 0 of } v(p, w) \end{aligned}$$

For $\alpha > 0$

$$\begin{aligned} v(\alpha p, \alpha w) &= v(\alpha p, \alpha e(p, u)) \\ &= v(\alpha p, e(\alpha p, u)) \text{ by homogeneity of degree 1 of } e(p, u) \text{ in } p \\ &= u \text{ by duality} \\ &= v(p, w) \text{ by duality} \end{aligned}$$

Part (ii,a) If the indirect utility function is strictly increasing in w and $v(p, w)$ is continuous in p, w then the expenditure function is strictly increasing in u .

Suppose that $u' > u$, and that $u = v(p, w)$ for some price level p .

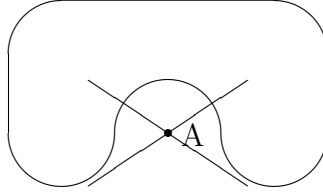
Part (ii,b) If the indirect utility function is strictly increasing in w and $v(p, w)$ is continuous in p, w then the expenditure function is strictly increasing in u .

Question 3.F.1. Prove formally that a closed, convex set $K \subset \mathbb{R}^L$ equals the intersection of half-spaces that contain it (use the separating hyperplane theorem).

Answer 3.F.1. Obviously, $K \subseteq \cap_{\alpha \in A} H_\alpha$ where A is some index on the half-spaces containing K , and the H_α are the half-spaces containing K . Now, suppose $\cap_{\alpha \in A} H_\alpha \not\subseteq K$, i.e. there exists some $\bar{x} \in \cap_{\alpha \in A} H_\alpha$ but $\bar{x} \notin K$. Since $\bar{x} \notin K$, by the separating hyperplane theorem, there exist $w \in \mathbb{R}$ and $p \in \mathbb{R}^L$ determining a hyperplane $p \cdot x = w$ such that $p \cdot \bar{x} < w$ and $p \cdot x \geq w$ for all $x \in K$. However, now we have constructed a half-space $\{x \in \mathbb{R}^L | p \cdot x \geq w\}$ containing K that does not contain $\bar{x}$, i.e. $\bar{x} \notin \cap_{\alpha \in A} H_\alpha$, a contradiction. Therefore, $\cap_{\alpha \in A} H_\alpha \subseteq K$. Since also $K \subseteq \cap_{\alpha \in A} H_\alpha$, $K = \cap_{\alpha \in A} H_\alpha$.

Question 3.F.2. Show by means of a graphic example that the separating hyperplane theorem does not hold for nonconvex sets. Then argue that if K is closed and not convex, there is always some $x \notin K$ that cannot be separated from K .

Answer 3.F.2. Here is a picture of a nonconvex set and a point that cannot be separated from it with a hyperplane.



Suppose K is closed and not convex. Then there exist $\bar{x}, \bar{y} \in K$ and $\alpha \in (0, 1)$ such that $\bar{z} = \alpha\bar{x} + (1 - \alpha)\bar{y} \notin K$. We claim that $\bar{z}$ cannot be separated from K with a hyperplane.

Suppose $\bar{z}$ could be separated from K with a hyperplane, so that there exist $w \in \mathbb{R}$ and $p \in \mathbb{R}^L$ determining a hyperplane $p \cdot x = w$ such that $p \cdot \bar{z} < w$ and $p \cdot x \geq w$ for all $x \in K$. Then, in particular, $p \cdot \bar{x} \geq w$ and $p \cdot \bar{y} \geq w$. Then for $\alpha \in (0, 1)$

$$\begin{aligned} \alpha p \cdot \bar{x} + (1 - \alpha) p \cdot \bar{y} &\geq \alpha w + (1 - \alpha) w \\ p \cdot (\alpha \cdot \bar{x} + (1 - \alpha) \cdot \bar{y}) &\geq w \\ p \cdot \bar{z} &\geq w \end{aligned}$$

which contradicts the assumption that $p \cdot \bar{z} < w$, so $\bar{z}$ and K cannot be separated by any hyperplane.

Question 3.G.1. Prove that Proposition 3.G.1 is implied by Roy's identity (Proposition 3.G.4).

Answer 3.G.1.

Roy's identity states

$$\begin{aligned} x(p, w) &= -\frac{\nabla_p v(p, w)}{\nabla_w v(p, w)} \\ 0 &= \nabla_p v(p, w) + x(p, w) \nabla_w v(p, w) \end{aligned}$$

Since $w = e(p, v(p, w)) = e(p, u)$ where $u = v(p, w)$

$$0 = \nabla_p v(p, e(p, u)) + x(p, e(p, u)) \nabla_w v(p, e(p, u))$$

Differentiating $v(p, e(p, u)) = u$ with respect to p

$$0 = \nabla_p v(p, e(p, u)) + \nabla_w v(p, e(p, u)) \nabla_p e(p, u)$$

Setting the previous two equations equal to each other

$$\begin{aligned} \nabla_p v(p, e(p, u)) + x(p, e(p, u)) \nabla_w v(p, e(p, u)) &= \nabla_p v(p, e(p, u)) + \nabla_w v(p, e(p, u)) \nabla_p e(p, u) \\ x(p, e(p, u)) &= \nabla_p e(p, u) \\ h(p, u) &= \nabla_p e(p, u) \end{aligned}$$

Question 3.G.2. Verify for the case of a Cobb-Douglas utility function that all of the propositions in Section 3.G hold.

Answer 3.G.2. First we need to find $x(p, w)$, $v(p, w)$, $h(p, u)$, and $e(p, u)$. Recall that Cobb-Douglas utility is $u(x) = \sum_{m=1}^L \alpha_m \ln x_m$ and without loss of generality, we assume that $\sum_{m=1}^L \alpha_m = 1$. Recall also

$$\begin{aligned} x(p, w) &= (\alpha_1 w / p_1, \dots, \alpha_L w / p_L) \\ v(p, w) &= \sum_{m=1}^L \alpha_m \ln(\alpha_m w / p_m) \end{aligned}$$

If the utility level is u , then

$$\begin{aligned}
u &= \sum_{m=1}^L \alpha_m \ln(\alpha_m e(p, u)/p_m) \\
\sum_{m=1}^L \alpha_m \ln e(p, u) &= u - \sum_{m=1}^L \alpha_m \ln(\alpha_m/p_m) \\
\ln e(p, u) &= u - \sum_{m=1}^L \alpha_m \ln(\alpha_m/p_m) \\
e(p, u) &= \exp \left\{ u - \sum_{m=1}^L \alpha_m \ln(\alpha_m/p_m) \right\} \\
h(p, u) &= x(p, e(p, u)) \\
h(p, u) &= (\alpha_1 e(p, u)/p_1, \dots, \alpha_L e(p, u)/p_L)
\end{aligned}$$

Now we can verify all the propositions. Proposition 3.G.1:

$$\begin{aligned}
\frac{\partial e(p, u)}{\partial p_l} &= \left[\frac{\partial}{\partial p_l} e(p, u) \right] e(p, u) \\
&= \alpha_l e(p, u)/p_l \\
&= h_l(p, u)
\end{aligned}$$

Proposition 3.G.2.i follows immediately from Proposition 3.G.1 so there is no need to differentiate both sides again. Proposition 3.G.2.ii immediately follows from the concavity of $e(p, u)$. Proposition 3.G.2.iii:

$$\begin{aligned}
\frac{\partial h_l(p, u)}{\partial p_k} &= \frac{\alpha_l \alpha_k}{p_l p_k} e(p, u) \\
\frac{\partial h_k(p, u)}{\partial p_l} &= \frac{\alpha_l \alpha_k}{p_l p_k} e(p, u)
\end{aligned}$$

Therefore

$$\frac{\partial h_l(p, u)}{\partial p_k} = \frac{\partial h_k(p, u)}{\partial p_l}$$

so $D_p h(p, u)$ is symmetric. Proposition 3.G.2.iv: The l th entry of $D_p h(p, u)p$ is

$$[D_p h(p, u)p]_l = \sum_{k=1}^L \frac{\partial h_l(p, u)}{\partial p_k} p_k$$

Using the derivatives that we compute to show Proposition 3.G.3

$$\begin{aligned}
&= \left[\frac{\alpha_l^2}{p_l^2} e(p, u) - \frac{\alpha_l}{p_l^2} e(p, u) \right] p_l + \sum_{k=1, k \neq l}^L \frac{\alpha_l \alpha_k}{p_l p_k} e(p, u) p_k \\
&= \left[\frac{\alpha_l^2}{p_l} e(p, u) - \frac{\alpha_l}{p_l} e(p, u) \right] + \sum_{k=1, k \neq l}^L \frac{\alpha_l \alpha_k}{p_l} e(p, u) \\
&= -\frac{\alpha_l}{p_l} e(p, u) + \sum_{k=1}^L \frac{\alpha_l \alpha_k}{p_l} e(p, u) \\
&= -\frac{\alpha_l}{p_l} e(p, u) + \frac{\alpha_l}{p_l} e(p, u) \sum_{k=1}^L p_k \\
&= 0
\end{aligned}$$

Proposition 3.G.3: For $l \neq k$

$$\begin{aligned}
\frac{\partial h_l(p, u)}{\partial p_k} &= \frac{\alpha_l \alpha_k}{p_l p_k} e(p, u) \\
\frac{\partial x_l(p, w)}{\partial p_k} &= 0 \\
\frac{\partial x_l(p, w)}{\partial w} x_k(p, w) &= \frac{\alpha_l}{p_l} \frac{\alpha_k w}{p_k}
\end{aligned}$$

Since $w = e(p, u)$

$$\frac{\partial h_l(p, u)}{\partial p_k} = \frac{\partial x_l(p, w)}{\partial p_k} + \frac{\partial x_l(p, w)}{\partial w} x_k(p, w)$$

Proposition 3.G.3: For $l = k$

$$\begin{aligned}
\frac{\partial h_l(p, u)}{\partial p_l} &= \frac{\alpha_l}{p_l} \left[\frac{\partial}{\partial p_l} e(p, u) \right] - \frac{\alpha_l}{p_l^2} e(p, u) \\
&= \frac{\alpha_l^2}{p_l^2} e(p, u) - \frac{\alpha_l}{p_l^2} e(p, u) \\
\frac{\partial x_l(p, w)}{\partial p_l} &= -\frac{\alpha_l w}{p_l^2} \\
\frac{\partial x_l(p, w)}{\partial w} x_l(p, w) &= \frac{\alpha_l}{p_l} \frac{\alpha_l w}{p_l}
\end{aligned}$$

Since $w = e(p, u)$

$$\frac{\partial h_l(p, u)}{\partial p_l} = \frac{\partial x_l(p, w)}{\partial p_l} + \frac{\partial x_l(p, w)}{\partial w} x_l(p, w)$$

Proposition 3.G.4:

$$\begin{aligned} \frac{\partial v(p, w)}{\partial p_l} &= -\alpha_l/p_l \\ \frac{\partial v(p, w)}{\partial w} &= \sum_{m=1}^L \frac{\alpha_m}{w} \\ &= \frac{1}{w} \\ -\frac{\partial v(p, w)/\partial p_l}{\partial v(p, w)/\partial w} &= \alpha_l w/p_l \\ &= x_l(p, w) \end{aligned}$$

Question 3.G.6. (F. M. Fisher) A consumer in a three-good economy (goods denoted x_1, x_2, x_3 ; prices denoted p_1, p_2, p_3) with wealth level $w > 0$ has demand functions for commodities 1 and 2 given by

$$\begin{aligned} x_1 &= 100 - 5\frac{p_1}{p_3} + \beta\frac{p_2}{p_3} + \delta\frac{w}{p_3} \\ x_2 &= \alpha + \beta\frac{p_1}{p_3} + \gamma\frac{p_2}{p_3} + \delta\frac{w}{p_3} \end{aligned}$$

where Greek letters are nonzero constants.

Question 3.G.6.a. Indicate how to calculate the demand for good 3 (but do not actually do it).

Answer 3.G.6.a. Substitute the demands for x_1 and x_2 into the consumer's budget constraint and solve for x_3 since the budget constraint holds with equality due to Walras' law.

Question 3.G.6.b. Are the demand functions for x_1 and x_2 appropriately homogeneous?

Answer 3.G.6.b. Yes, by inspection both are homogeneous of degree 0 in (p, w) .

Question 3.G.7. A striking duality is obtained by using the concept of *indirect demand function*. Fix w at some level, say $w = 1$; from now on, we write $x(p, 1) = x(p)$, $v(p, 1) = v(p)$. The *indirect demand function* $g(x)$ is the inverse of $x(p)$; that

is, it is the rule that assigns to every commodity bundle $x \gg 0$ the price vector $g(x)$ such that $x = x(g(x), 1)$. Show that

$$g(x) = \frac{1}{x \cdot \nabla u(x)} \nabla u(x)$$

Deduce from Proposition 3.G.4 that

$$x(p) = \frac{1}{p \cdot \nabla v(p)} \nabla v(p)$$

Answer 3.G.7. First, show that

$$g(x) = \frac{1}{x \cdot \nabla u(x)} \nabla u(x)$$

The first order conditions from the UMP are $\lambda p_l = \frac{\partial u}{\partial x_l}$ for all $l = 1, \dots, L$, or, in vector notation

$$\lambda p = \nabla u(x)$$

Dotting this equation by x on both sides and applying Walras' law ($p \cdot x(p) = w = 1$)

$$\lambda = x \cdot \nabla u(x)$$

Dividing these two expressions

$$\frac{\lambda p}{\lambda} = \frac{\nabla u(x)}{x \cdot \nabla u(x)}$$

Since $p = g(x)$ by definition

$$g(x) = \frac{1}{x \cdot \nabla u(x)} \nabla u(x)$$

Similarly, we show that

$$x(p) = \frac{1}{p \cdot \nabla v(p)} \nabla v(p)$$

By Roy's identity

$$x(p) = -\frac{\nabla_p v(p)}{\nabla_w v(p)}$$

Dotting on both sides by p and applying Walras' law ($p \cdot x(p) = w = 1$)

$$1 = -\frac{p \cdot \nabla_p v(p)}{\nabla_w v(p)}$$

Dividing these two equations

$$x(p) = \frac{1}{p \cdot \nabla v(p)} \nabla v(p)$$

Question 3.G.8. The indirect utility function $v(p, w)$ is logarithmically homogeneous if $v(p, \alpha w) = v(p, w) + \ln \alpha$ for $\alpha > 0$ [in other words, $v(p, w) = \ln(v^*(p, w))$, where $v^*(p, w)$ is homogeneous of degree one]. Show that if $v(\cdot, \cdot)$ is logarithmically homogeneous, then $x(p, 1) = -\nabla_p v(p, 1)$.

Answer 3.G.8. By Roy's identity

$$x(p, 1) = -\frac{\nabla_p v(p, 1)}{\nabla_w v(p, 1)}$$

Now it is sufficient to show that $\nabla_w v(p, 1) = 1$.

$$\nabla_w v(p, 1) = \left. \frac{\partial}{\partial w} v(p, w) \right|_{w=1}$$

Since $v(\cdot, \cdot)$ is logarithmically homogeneous

$$= \left. \frac{\partial}{\partial w} (v(p, 1) + \ln w) \right|_{w=1}$$

Differentiating and evaluating at $w = 1$

$$\begin{aligned} &= 0 + 1 \\ &= 1 \end{aligned}$$

Question 3.G.9. Compute the Slutsky matrix from the indirect utility function

Answer 3.G.9. The Slutsky matrix is the $L \times L$ matrix with lk entry

$$s_{lk} = \frac{\partial x_l(p, w)}{\partial p_k} + \frac{\partial x_l(p, w)}{\partial w} x_k(p, w)$$

Roy's identity gives the relationship between $x(p, w)$ and $v(p, w)$ and we will use it to express s_{lk} in terms of $v(p, w)$. To keep the notation clean, we will drop the

arguments of $x(p, w)$ and $v(p, w)$, i.e. we will simply write x and v . Also, we will write x_l to denote the demand for good l , but v_l to denote $\frac{\partial v}{\partial p_l}$ and v_w for $\frac{\partial v}{\partial w}$

Roy's identity is

$$x_l = -\frac{v_l}{v_w}$$

Differentiating with respect to p_k

$$\begin{aligned}\frac{\partial x_l}{\partial p_k} &= -\frac{v_{lk}}{v_w} + \frac{v_l v_{wk}}{v_w^2} \\ &= \frac{v_l v_{wk} - v_{lk} v_w}{v_w^2}\end{aligned}$$

Differentiating with respect to w

$$\begin{aligned}\frac{\partial x_l}{\partial w} &= -\frac{v_{lw}}{v_w} + \frac{v_l v_{ww}}{v_w^2} \\ &= \frac{v_l v_{ww} - v_{lw} v_w}{v_w^2}\end{aligned}$$

Substituting to find s_{lk} in terms of v

$$s_{lk} = \frac{v_l v_{wk} - v_{lk} v_w}{v_w^2} - \frac{v_l v_{ww} - v_{lw} v_w}{v_w^2} \frac{v_k}{v_w}$$

Question 3.G.10. For a function of the Gorman form $v(p, w) = a(p) + b(p)w$, which properties will the functions $a(\cdot)$ and $b(\cdot)$ have to satisfy for $v(p, w)$ to qualify as an indirect utility function?

Answer 3.G.10. In order for $v(p, w)$ to be homogeneous of degree 0, we need $a(\cdot)$ homogeneous of degree 0 and $b(\cdot)$ homogeneous of degree -1 . For $v(p, w)$ to be strictly increasing in w , we need $b(p) > 0$. For $v(p, w)$ to be non-increasing in p_l for all l , we need $a_l(p) + b_l(p)w \leq 0$ for all (p, w) . Finally, for $v(p, w)$ to be quasiconvex, we require $a(\cdot)$ and $b(\cdot)$ to be quasiconvex.

Question 3.G.11. Verify that an indirect utility function in Gorman form exhibits linear wealth-expansion curves.

Answer 3.G.11.

By Roy's identity

$$\begin{aligned} x(p, w) &= -\frac{\nabla_p v(p, w)}{\nabla_w v(p, w)} \\ &= -\frac{\nabla_p(a(p) + b(p)w)}{b(p)} \\ \frac{\partial x(p, w)}{\partial w} &= -\frac{\nabla_p b(p)}{b(p)} \end{aligned}$$

which is constant for any fixed p . (Recall that wealth-expansion considers a fixed price level while wealth varies.)

Question 3.G.12. What restrictions on the Gorman form correspond to the cases of homothetic and quasilinear preferences?

Answer 3.G.12. From Exercise 3.D.3, preferences are homothetic if and only if $v(p, w)$ is homogeneous of degree 1 in w . If $v(p, w)$ also has Gorman form then for every $\alpha > 0$

$$\begin{aligned} v(p, \alpha w) &= \alpha v(p, w) \\ a(p) + \alpha b(p)w &= \alpha a(p) + \alpha b(p)w \\ a(p) &= \alpha a(p) \end{aligned}$$

Since $\alpha > 0$, this last equation is only true when $a(p) = 0$ for all p . Additionally, the Gorman form must satisfy the requirements described in Exercise 3.G.10.

From Exercise 3.D.4, preferences are quasilinear if and only if $v(p, w) = w + \phi(p)$. If $v(p, w)$ also has Gorman form then

$$w + \phi(p) = a(p) + b(p)w$$

Since this last equation must hold for all p , we conclude that $b(p) = 1$ for all p . Additionally, the Gorman form must satisfy the requirements described in Exercise 3.G.10.

Question 3.G.13. Suppose that the indirect utility function $v(p, w)$ is a polynomial of degree n on w (with coefficients that may depend on p). Show that any individual wealth expansion path is contained in a linear subspace of at most dimension $n + 1$. Interpret.

Question 3.G.13. The problem tells us that there are functions $a_0(p), a_1(p), \dots, a_n(p)$ such that

$$v(p, w) = a_0(p) + a_1(p)w + \dots + a_n(p)w^n$$

Applying Roy's identity

$$x(p, w) = -\frac{\sum_{k=0}^n \nabla_p a_k(p) w^k}{\sum_{m=1}^n m a_m(p) w^{m-1}}$$

$$x(p, w) = -\sum_{k=0}^n \frac{w^k}{\sum_{m=1}^n m a_m(p) w^{m-1}} \nabla_p a_k(p)$$

Now it is easy to see for a fixed price level that this sum is a linear combination of elements of $\{\nabla_p a_k(p) | k = 0, \dots, n\}$ with coefficients depending on w . Since there are $n + 1$ elements in this set, they span a space of dimension at most $n + 1$. Thus, any individual wealth expansion path is contained in a linear subspace of at most dimension $n + 1$.

Question 3.G.14. The matrix below records the (Walrasian) demand substitution effects for a consumer endowed with rational preferences and consuming three goods at the prices $p_1 = 1, p_2 = 2$, and $p_3 = 6$:

$$\begin{bmatrix} -10 & a & b \\ c & -4 & d \\ 3 & e & f \end{bmatrix}$$

Supply the missing numbers. Does the resulting matrix possess all the properties of a substitution matrix?

Answer 3.G.14. Additionally, assume that preferences are strictly convex, continuous, and locally non-satiated, so that we can apply Proposition 3.G.2. Adding an assumption is generally a poor way of solving a problem, but I could not come up with another way, and in fact the “official” solutions make this assumption as well.

By Proposition 3.G.2, the substitution matrix is symmetric, so $b = 3$, $a = c$, and $d = e$. Further, by Proposition 3.G.2, $Sp = 0$.

$$\begin{bmatrix} -10 & a & 3 \\ a & -4 & d \\ 3 & d & f \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Therefore

$$0 = -10 + 2a + 18$$

$$0 = a - 8 + 6d$$

$$0 = 3 + 2e + 6f$$

Solving

$$a = -4$$

$$d = 2$$

$$f = -7/6$$

So the complete substitution matrix is

$$\begin{bmatrix} -10 & -4 & 3 \\ -4 & -4 & 2 \\ 3 & 2 & -7/6 \end{bmatrix}$$

Now we will check that this matrix is negative semidefinite to make sure that it is negative semidefinite. Obviously, one eigenvalue is 0 since $Sp = 0$. If we can show that the other two eigenvalues are not positive, then we will be done.

$$\begin{aligned} 0 &= \begin{vmatrix} -10 - \lambda & -4 & 3 \\ -4 & -4 - \lambda & 2 \\ 3 & 2 & -7/6 - \lambda \end{vmatrix} \\ &= (-10 - \lambda)[(-4 - \lambda)(-7/6 - \lambda) - 4] + 4[-4(-7/6 - \lambda) - 6] + 3[-8 - 3(-4 - \lambda)] \end{aligned}$$

$$0 = \lambda^3 + 82\lambda^2/3 + 272\lambda/6$$

$$\lambda = 0 \text{ or } \frac{-164 \pm \sqrt{164^2 - (4)(6)(272)}}{(2)(6)} < 0$$

Question 3.G.15. Consider the utility function

$$u = 2x_1^{1/2} + 4x_2^{1/2}$$

Question 3.G.15.a. Find the demand functions for goods 1 and 2 as they depend on prices and wealth.

Answer 3.G.15.a. Solve

$$\max_{x_1, x_2} 2x_1^{1/2} + 4x_2^{1/2} \text{ subject to } p_1x_1 + p_2x_2 \leq w$$

The preferences are locally nonsatiated, so the constraint binds. The above constrained maximization is equivalent to

$$\max_{x_1, x_2} 2x_1^{1/2} + 4x_2^{1/2} + \lambda(w - p_1x_1 - p_2x_2)$$

The optimal x_1, x_2 satisfy

$$\begin{aligned} 0 &= x_1^{-1/2} - \lambda p_1 \\ 0 &= 2x_2^{-1/2} - \lambda p_2 \\ 0 &= w - p_1x_1 - p_2x_2 \end{aligned}$$

Solving simultaneously

$$\begin{aligned} x_1(p, w) &= \frac{wp_2}{4p_1^2 + p_1p_2} \\ x_2(p, w) &= \frac{4wp_1}{4p_1p_2 + p_2^2} \end{aligned}$$

Question 3.G.15.b. Find the compensated demand function $h(\cdot)$.

Answer 3.G.15.b. Solve

$$\min_{x_1, x_2} p_1x_1 + p_2x_2 \text{ subject to } 2x_1^{1/2} + 4x_2^{1/2} \geq u$$

The preferences are locally nonsatiated, so the constraint binds. The above constrained minimization is equivalent to

$$\min_{x_1, x_2} p_1x_1 + p_2x_2 - \lambda(2x_1^{1/2} + 4x_2^{1/2} - u)$$

The optimal x_1, x_2 satisfy

$$\begin{aligned} 0 &= p_1 - \lambda x_1^{-1/2} \\ 0 &= p_2 - 2\lambda x_2^{-1/2} \\ 0 &= 2x_1^{1/2} + 4x_2^{1/2} - u \end{aligned}$$

Solving simultaneously

$$\begin{aligned} h_1(p, u) &= \left(\frac{p_2u}{8p_1 + 2p_2} \right)^2 \\ h_2(p, u) &= \left(\frac{2p_1u}{8p_1 + 2p_2} \right)^2 \end{aligned}$$

Question 3.G.15.c. Find the expenditure function, and verify that $h(p, u) = \nabla_p e(p, u)$.

Answer 3.G.15.c. Find $e(p, u)$

$$\begin{aligned} e(p, u) &= p_1 \left(\frac{p_2 u}{8p_1 + 2p_2} \right)^2 + p_2 \left(\frac{2p_1 u}{8p_1 + 2p_2} \right)^2 \\ &= \frac{p_1 p_2^2 u^2 + 4p_1^2 p_2 u^2}{(8p_1 + 2p_2)^2} \end{aligned}$$

Show that $\partial e(p, u)/\partial p_1 = h_1(p, u)$

$$\begin{aligned} \frac{\partial e(p, u)}{\partial p_1} &= \frac{p_2^2 u^2 + 8p_1 p_2 u^2}{(8p_1 + 2p_2)^2} - 16 \frac{p_1 p_2^2 u^2 + 4p_1^2 p_2 u^2}{(8p_1 + 2p_2)^3} \\ &= p_2 u^2 \left\{ \frac{(p_2 + 8p_1)(2p_2 + 8p_1) - 16p_1 p_2 - 64p_1^2}{(8p_1 + 2p_2)^3} \right\} \\ &= p_2^2 u^2 \left\{ \frac{8p_1 + 2p_2}{(8p_1 + 2p_2)^3} \right\} \\ &= h_1(p, u) \\ \frac{\partial e(p, u)}{\partial p_1} &= \frac{2p_1 p_2 u^2 + 4p_1^2 u^2}{(8p_1 + 2p_2)^2} - 4 \frac{p_1 p_2^2 u^2 + 4p_1^2 p_2 u^2}{(8p_1 + 2p_2)^3} \\ &= p_1 u^2 \left\{ \frac{(2p_2 + 4p_1)(8p_1 + 2p_2) - 4p_2^2 - 16p_1 p_2}{(8p_1 + 2p_2)^3} \right\} \\ &= p_1^2 u^2 \left\{ \frac{4(8p_1 + 2p_2)}{(8p_1 + 2p_2)^3} \right\} \\ &= h_2(p, u) \end{aligned}$$

Question 3.G.15.d. Find the indirect utility function and verify Roy's identity.

Answer 3.G.15.d. The indirect utility function is

$$\begin{aligned} v(p, w) &= u(x_1(p, w), x_2(p, w)) \\ &= 2 \left(\frac{wp_2}{4p_1^2 + p_1 p_2} \right)^{1/2} + 4 \left(\frac{4wp_1}{4p_1 p_2 + p_2^2} \right)^{1/2} \\ &= \frac{2\sqrt{w}\sqrt{4p_1 + p_2}}{\sqrt{p_1 p_2}} \end{aligned}$$

To verify Roy's identity, first calculate the derivatives of $v(p, w)$:

$$\begin{aligned}\frac{\partial v(p, w)}{\partial p_1} &= \frac{-p_2^2 \sqrt{w}}{(p_1 p_2 (4p_1 + p_2))^{3/2}} \\ \frac{\partial v(p, w)}{\partial p_2} &= \frac{-4p_1^2 \sqrt{w}}{(p_1 p_2 (4p_1 + p_2))^{3/2}} \\ \nabla_w v(p, w) &= \frac{\sqrt{4p_1 + p_2}}{\sqrt{w} \sqrt{p_1 p_2}}\end{aligned}$$

For Roy's identity to hold, we require $x(p, w) = -\nabla_p v(p, w) / \nabla_w v(p, w)$. Substituting the above expressions

$$\begin{aligned}x_1(p, w) &= -\frac{\frac{-p_2^2 \sqrt{w}}{(p_1 p_2 (4p_1 + p_2))^{3/2}}}{\frac{\sqrt{4p_1 + p_2}}{\sqrt{w} \sqrt{p_1 p_2}}} \\ &= \frac{p_2 w}{(4p_1 + p_2)^2 p_1} \\ x_2(p, w) &= -\frac{\frac{-4p_1^2 \sqrt{w}}{(p_1 p_2 (4p_1 + p_2))^{3/2}}}{\frac{\sqrt{4p_1 + p_2}}{\sqrt{w} \sqrt{p_1 p_2}}} \\ &= \frac{4p_1 w}{(4p_1 + p_2) p_2}\end{aligned}$$

Question 3.H.2. Use Proposition 3.F.1 to argue that if $e(p, u)$ is differentiable in p , then there are no (strongly monotone) nonconvex preferences generating $e(\cdot)$.

Answer 3.H.2.

Question 3.H.3. How would you recover $v(p, w)$ from $e(p, u)$?

Answer 3.H.3.

$$\begin{aligned}\nabla_p e(p, u) &= h(p, u) \\ &= x(p, e(p, u)) \\ &= -\frac{\nabla_p v(p, w)}{\nabla_w v(p, w)} \text{ where } w = e(p, u)\end{aligned}$$

Now solve the partial differential equation by differentiating again with respect to p and using the symmetry of the Slutsky matrix.

“Official” solution is much better: remember that $e(p, u)$ is strictly increasing in u and therefore invertible in u . Then $e(p, v(p, w)) = w$ which implies $v(p, w) = e^{-1}(p, w)$.

Question 3.H.5. Suppose you know the indirect utility function. How would you recover from it the expenditure function and the direct utility function?

Answer 3.H.5. If $w = e(p, u)$, then $u = v(p, e(p, u))$. Since $v(p, w)$ is strictly increasing in w , it is invertible in w , so $e(p, u) = v^{-1}(p, u)$. To recover the direct utility function, extract $x(p, w)$. Then $u(x) = u(x(p, w)) = v(p, w)$.

Question 3.H.6. Suppose that you observe the Walrasian demand functions $x_l(p, w) = \alpha_l w / p_l$ for all $l = 1, \dots, L$ with $\sum_l \alpha_l = 1$. Derive the expenditure function of this demand system. What is the consumer’s utility function?

Answer 3.H.6. First, we need to show that $e(p, u)$ and $u(x)$ can be recovered at all, i.e. we need to show that the Slutsky matrix is symmetric. Recall that for all $l = 1, \dots, L$

$$\frac{\partial e(p, u)}{\partial p_l} = h_l(p, u) = x_l(p, e(p, u)) = \alpha_l e(p, u) / p_l$$

To show that the Slutsky matrix is symmetric, we want to show that $\frac{\partial^2 e(p, u)}{\partial p_k \partial p_l} = \frac{\partial^2 e(p, u)}{\partial p_l \partial p_k}$ for all $k \neq l$. Differentiating

$$\begin{aligned} \frac{\partial^2 e(p, u)}{\partial p_k \partial p_l} &= \frac{\alpha_l}{p_l} \frac{\partial e(p, u)}{\partial p_k} \\ &= \frac{\alpha_l \alpha_k e(p, u)}{p_l p_k} \\ \frac{\partial^2 e(p, u)}{\partial p_l \partial p_k} &= \frac{\alpha_k}{p_k} \frac{\partial e(p, u)}{\partial p_l} \\ &= \frac{\alpha_k \alpha_l e(p, u)}{p_k p_l} \end{aligned}$$

Question 3.I.1. Suppose that the change from price vector p^0 to price vector p^1 involves a change in the prices of good 1 (from p_1^0 to p_1^1) and good 2 (from p_2^0 to p_2^1). Express the equivalent variation in terms of the sum of integrals under appropriate Hicksian demand curves for goods 1 and 2. Do the same for the compensating variation measure. Show also that if there are no wealth effects for either good, the compensating and equivalent variations are equal.

Answer 3.I.1. Let $p = (p_3, p_4, \dots, p_L)$ be the prices of goods 3, 4, $\dots$, L . Then when u^0 and u^1 are utility levels such that $w = e(p^0, u^0) = e(p^1, u^1)$

$$\begin{aligned}
 EV(p^0, p^1, w) &= e(p^0, u^1) - e(p^0, u^0) \\
 &= e(p^0, u^1) - e(p_1^1, p_2^0, p, u^1) + e(p_1^1, p_2^0, p, u^1) - e(p^1, u^1) \\
 &= e(p_1^0, p_2^0, p, u^1) - e(p_1^1, p_2^0, p, u^1) + e(p_1^1, p_2^0, p, u^1) - e(p_1^1, p_2^1, p, u^1) \\
 &= \int_{p_1^1}^{p_1^0} h_1(p_1, p_2^0, p, u^1) dp_1 + \int_{p_2^1}^{p_2^0} h_2(p_1^1, p_2, p, u^1) dp_2 \\
 CV(p^0, p^1, w) &= e(p^1, u^1) - e(p^1, u^0) \\
 &= e(p^0, u^0) - e(p_1^1, p_2^0, p, u^0) + e(p_1^1, p_2^0, p, u^0) - e(p^1, u^0) \\
 &= e(p_1^0, p_2^0, p, u^0) - e(p_1^1, p_2^0, p, u^0) + e(p_1^1, p_2^0, p, u^0) - e(p_1^1, p_2^1, p, u^0) \\
 &= \int_{p_1^1}^{p_1^0} h_1(p_1, p_2^0, p, u^0) dp_1 + \int_{p_2^1}^{p_2^0} h_2(p_1^1, p_2, p, u^0) dp_2
 \end{aligned}$$

If there are no wealth effects for goods 1 and 2 then

$$\begin{aligned}
 EV(p^0, p^1, w) &= \int_{p_1^1}^{p_1^0} h_1(p_1, p_2^0, p, u^1) dp_1 + \int_{p_2^1}^{p_2^0} h_2(p_1^1, p_2, p, u^1) dp_2 \\
 &= \int_{p_1^1}^{p_1^0} x_1(p_1, p_2^0, p, w) dp_1 + \int_{p_2^1}^{p_2^0} x_2(p_1^1, p_2, p, w) dp_2 \\
 CV(p^0, p^1, w) &= \int_{p_1^1}^{p_1^0} h_1(p_1, p_2^0, p, u^0) dp_1 + \int_{p_2^1}^{p_2^0} h_2(p_1^1, p_2, p, u^0) dp_2 \\
 &= \int_{p_1^1}^{p_1^0} x_1(p_1, p_2^0, p, w) dp_1 + \int_{p_2^1}^{p_2^0} x_2(p_1^1, p_2, p, w) dp_2
 \end{aligned}$$

Question 3.I.2. Calculate the derivative of the deadweight loss measures (3.I.5) and (3.I.6) with respect to t . Show that, evaluated at $t = 0$, these derivatives are equal to zero, but that if $h_1(p, u)$ is strictly decreasing in p_1 , they are strictly positive at all $t > 0$. Interpret.

Answer 3.I.2. Equation 3.I.5:

$$\begin{aligned}
\frac{\partial}{\partial t} [-T - EV(p^0, p^1, w)] &= \frac{\partial}{\partial t} \left[\int_{p_1^0}^{p_1^0+t} h_1(p_1, \bar{p}_{-1}, u^1) - h_1(p_1^0 + t, \bar{p}_{-1}, u^1) dp_1 \right] \\
&= \frac{\partial}{\partial t} \left[\int_{p_1^0}^{p_1^0+t} h_1(p_1, \bar{p}_{-1}, u^1) dp_1 - t h_1(p_1^0 + t, \bar{p}_{-1}, u^1) \right] \\
&= h_1(p_1^0 + t, \bar{p}_{-1}, u^1) - h_1(p_1^0 + t, \bar{p}_{-1}, u^1) - t \frac{\partial h_1(p_1^0 + t, \bar{p}_{-1}, u^1)}{\partial p_1} \\
&= -t \frac{\partial h_1(p_1^0 + t, \bar{p}_{-1}, u^1)}{\partial p_1}
\end{aligned}$$

Equation 3.I.6

$$\begin{aligned}
\frac{\partial}{\partial t} [-CV(p^0, p^1, w) - t h_1(p^1, u^0)] &= \frac{\partial}{\partial t} \left[\int_{p_1^0}^{p_1^0+t} h_1(p_1, \bar{p}_{-1}, u^0) - h_1(p_1^0 + t, \bar{p}_{-1}, u^0) dp_1 \right] \\
&= \frac{\partial}{\partial t} \left[\int_{p_1^0}^{p_1^0+t} h_1(p_1, \bar{p}_{-1}, u^0) dp_1 - t h_1(p_1^0 + t, \bar{p}_{-1}, u^0) \right] \\
&= h_1(p_1^0 + t, \bar{p}_{-1}, u^0) - h_1(p_1^0 + t, \bar{p}_{-1}, u^0) - t \frac{\partial h_1(p_1^0 + t, \bar{p}_{-1}, u^0)}{\partial p_1} \\
&= -t \frac{\partial h_1(p_1^0 + t, \bar{p}_{-1}, u^0)}{\partial p_1}
\end{aligned}$$

In both cases, if $t = 0$, the derivative is 0. If $h_1(p, u)$ is strictly decreasing in p_1 , then $\frac{\partial h_1(p_1^0+t, \bar{p}_{-1}, u)}{\partial p_1} < 0$, for $u = u^0, u^1$, so $-t \frac{\partial h_1(p_1^0+t, \bar{p}_{-1}, u)}{\partial p_1} > 0$ whenever $t > 0$. Intuitively, when there is no commodity tax, there is no resulting loss. If there is a commodity tax, the loss increases with the tax rate.

Question 3.I.5. Show that if $u(x)$ is quasilinear with respect to the first good (and we fix $p_1 = 1$), then $CV(p^0, p^1, w) = EV(p^0, p^1, w)$ for any (p^0, p^1, w) .

Answer 3.I.5. (From Laura Rosner)

By e_1 denote the $L \times 1$ vector with 1 in the first entry and 0 everywhere else.

$$\begin{aligned}
CV(p^0, p^1, w) &= e(p^1, u^1) - e(p^1, u^0) \\
&= p^1 \cdot [h(p^1, u^1) - h(p^1, u^0)]
\end{aligned}$$

By Exercise 3.E.7

$$\begin{aligned}
 &= p^1 \cdot (u^1 - u^0)e_1 \\
 EV(p^0, p^1, w) &= e(p^0, u^1) - e(p^0, u^0) \\
 &= p^0 \cdot [h(p^0, u^1) - h(p^0, u^0)]
 \end{aligned}$$

By Exercise 3.E.7

$$= p^0 \cdot (u^1 - u^0)e_1$$

Since $p_1 = 1$ in both p^0 and p^1

$$CV(p^0, p^1, w) = EV(p^0, p^1, w) = u^1 - u^0$$

Question 3.I.7. There are three commodities (i.e. $L = 3$), of which the third is a numeraire (let $p_3 = 1$). The market demand function $x(p, w)$ has

$$\begin{aligned}
 x_1(p, w) &= a + bp_1 + cp_2 \\
 x_2(p, w) &= d + ep_1 + gp_2
 \end{aligned}$$

Question 3.I.7.a. Give the parameter restrictions implied by utility maximization.

Answer 3.I.7.a. We require that $x_1(p, w)$ and $x_2(p, w)$ be homogeneous of degree 0, i.e. $b = c = e = g = 0$, and that demand be non-negative, i.e. $a \geq 0$ and $d \geq 0$. Compute $x_3(p, w) = (w - ap_1 - dp_2)/p_3$ by Walras's law, which by inspection satisfies homogeneity of degree 0.

Question 3.I.7.b. Estimate the equivalent variation for a change of prices from $(p_1, p_2) = (1, 1)$ to $(p_1, p_2) = (2, 2)$. Verify that without appropriate symmetry, there is no path independence. Assume symmetry for the rest of the exercise.

Question 3.J.1. Show that when $L = 2$, $x(p, w)$ satisfies the strong axiom if and only if it satisfies the weak axiom.

Answer 3.J.1. Obviously (even for $L > 2$) the strong axiom implies the weak axiom, since application of the strong axiom to any list with $N = 2$ is exactly the statement of the weak axiom.

The other direction, that the weak axiom implies the strong axiom when $L = 2$ is trickier. Recall from Exercise 2.F.11 that if and $L = 2$, then the Slutsky matrix is symmetric, which as we learned in Section 3.H (page 80, no particular result) is a sufficient condition to recover rational preferences that induce demand $x(p, w)$. In turn, rational preferences imply that demand satisfies the strong axiom.

Question 3.AA.1.**Chapter 4**

Question 4.B.1. Prove the sufficiency part of Proposition 4.B.1. Show also that if preferences admit Gorman form indirect utility functions with the same $b(p)$, then preferences admit expenditure functions of the form $e_i(p, u_i) = c(p)u_i + d_i(p)$.

Answer 4.B.1. To show the sufficiency part of Proposition 4.B.1, apply Roy's identity to $v_i(p, w_i) = a_i(p) + b(p)w_i$.

$$\begin{aligned} x_i(p, w_i) &= -\frac{\nabla_p[a_i(p) + b(p)w_i]}{\nabla_{w_i}[a_i(p) + b(p)w_i]} \\ &= -\frac{\nabla_p a_i(p)}{b(p)} - \frac{\nabla_p b(p)}{b(p)}w_i \end{aligned}$$

At a fixed price level while w_i varies, this is the equation of a hyperplane with slope $-\frac{\nabla_p b(p)}{b(p)}$ independent of i . In other words, the wealth expansion paths are parallel lines.

Recall that if $e_i(p, u_i) = w_i$ then $v_i(p, e_i(p, u_i)) = u_i$. Writing v_i in Gorman form

$$u_i = a_i(p) + b(p)e_i(p, u_i)$$

Rearranging

$$e_i(p, u_i) = \frac{u_i}{b(p)} - \frac{a_i(p)}{b(p)}$$

With $c(p) = \frac{1}{b(p)}$ and $d_i(p) = -\frac{a_i(p)}{b(p)}$ we get the desired result.

Question 4.B.2. Suppose that there are I consumers and L commodities. Consumers differ only by their wealth levels w_i and by a taste parameter s_i , which we could call *family size*. Thus, denote the indirect utility function of consumer i by $v(p, w_i, s_i)$. The corresponding Walrasian demand function for consumer i is $x(p, w_i, s_i)$.

Question 4.B.2.a. Fix $(s_1, \dots, s_I)$. Show that if for any $(w_1, \dots, w_I)$ aggregate demand can be written as a function of only p and aggregate wealth $w = \sum_i w_i$ (or, equivalently, average wealth), and if every consumer's preference relationship $\succeq_i$ is homothetic, then all these preferences must be identical [and so $x(p, w_i, s_i)$ must be independent of s_i].

Answer 4.B.2.a. From Exercise 3.G.12, we know that for homothetic preferences, the indirect utility function has Gorman form $b_i(p)w_i$ (i.e. $a_i(p) = 0$). Since aggregate demand can be written as a function of prices and aggregate wealth, by Proposition 4.B.1 we know that the individual wealth expansion paths are parallel straight lines, i.e. $b_i(p) = b_j(p)$ for all $i, j = 1, \dots, I$. In other words, the indirect utility functions are identical, ergo the preferences are identical.

Question 4.B.2.b. Give a sufficient condition for aggregate demand to depend only on aggregate wealth w and $\sum_i s_i$ (or, equivalently, average wealth and average family size).

Answer 4.B.2.b. Presumably, this is a parallel of Proposition 4.B.1, namely that the indirect utilities for consumers i and j must satisfy

$$\frac{\partial v_i(p, w_i, s_i)}{\partial w_i} = \frac{\partial v_j(p, w_j, s_j)}{\partial w_j} \text{ and } \frac{\partial v_i(p, w_i, s_i)}{\partial s_i} = \frac{\partial v_j(p, w_j, s_j)}{\partial s_j}$$

for all $i, j = 1, \dots, I$.

Question 4.C.5. Prove Proposition 4.C.3. To that effect, you can fix $w = 1$. The proof is best done in terms of the indirect demand function $g_i(x) = (1/x \cdot \nabla u_i(x)) \nabla u_i(x)$ [note that $x_i = x(g_i(x), 1)$]. For an individual consumer, the ULD is self-dual; that is, it is equivalent to $(g_i(x) - g_i(y)) \cdot (x - y) < 0$ for all $x \neq y$. In turn, this property is implied by the negative definiteness of $Dg_i(x)$ for all x . Hence, concentrate on proving this last property. More specifically, let $v \neq 0$, and denote $q = \nabla u_i(x)$ and $C = D^2 u_i(x)$. You want to prove $v \cdot Dg_i(x)v < 0$. [Hint: You can first assume $q \cdot v = q \cdot x$; then differentiate $g_i(x)$, and make use of the equality $v \cdot Cv - x \cdot Cv = (v - \frac{1}{2}x) \cdot C(v - \frac{1}{2}x) - \frac{1}{4}x \cdot Cx$.]

Answer 4.C.5. Recall that the uncompensated law of demand says that

$$(p' - p) \cdot [x_i(p', w_i) - x_i(p, w_i)] \leq 0$$

for any p, p' , and w_i , with strict inequality if $x_i(p', w_i) \neq x_i(p, w_i)$.

Proposition 4.C.3: Suppose that $\succeq_i$ is defined on the consumption set $X = \mathbb{R}_+^L$ and is representable by a twice continuously differentiable concave function $u_i(\cdot)$. If

$$-\frac{x_i \cdot D^2 u_i(x_i) x_i}{x_i \cdot \nabla u_i(x_i)} < 4 \text{ for all } x_i$$

then $x_i(p, w_i)$ satisfies the unrestricted law of demand property.

Without loss of generality, fix $w_i = 1$. Recall from Exercise 3.G.7 that $g_i(x)$ is the price vector at which $x_i = x(g_i(x), 1)$. Writing the ULD in terms of the indirect

demand function, we want to show that for any two bundles x and y

$$(g(x) - g(y)) \cdot (x - y) \leq 0$$

Question 4.C.11. Suppose there are two consumers, 1 and 2, with utility functions over two goods, 1 and 2, of $u_1(x_{11}, x_{21}) = x_{11} + 4\sqrt{x_{21}}$ and $u_2(x_{12}, x_{22}) = 4\sqrt{x_{12}} + x_{22}$. The two consumers have identical wealth levels $w_1 = w_2 = w/2$.

Question 4.C.11.a. Calculate the individual demand functions and the aggregate demand function.

Answer 4.C.11.a.

Consumer 1's demand is

$$\operatorname{argmax}_{(x_{11}, x_{21})} x_{11} + 4\sqrt{x_{21}} \text{ subject to } p_1 x_1 + p_2 x_2 = w/2$$

FOCs

$$\begin{aligned} 0 &= 1 - \lambda p_1 \\ 0 &= \frac{2}{\sqrt{x_{21}}} - \lambda p_2 \end{aligned}$$

Solving

$$\begin{aligned} x_{11}(p, w) &= \frac{w}{2p_1} - \frac{4p_1}{p_2} \\ x_{21}(p, w) &= \frac{4p_1^2}{p_2^2} \end{aligned}$$

Symmetrically, consumer 2's demand is

$$\begin{aligned} x_{12}(p, w) &= \frac{4p_2^2}{p_1^2} \\ x_{22}(p, w) &= \frac{w}{2p_2} - \frac{4p_2}{p_1} \end{aligned}$$

Summing, aggregate demand is

$$x(p, w) = \left(\frac{w}{2p_1} - \frac{4p_1}{p_2} + \frac{4p_2^2}{p_1^2}, \frac{4p_1^2}{p_2^2} + \frac{w}{2p_2} - \frac{4p_2}{p_1} \right)$$

As a side note, this shows that aggregate demand does not preserve quasilinearity in different goods.

Question 4.C.11.b. Compute the individual Slutsky matrices $S_i(p, w/2)$ (for $i = 1, 2$) and the aggregate Slutsky matrix $S(p, w)$. [*Hint*: Note that for this two good example, only one element of each matrix must be computed to determine the entire matrix.] Show that $dp \cdot S(p, w)dp < 0$ all $dp \neq 0$ not proportional to p . Conclude that aggregate demand satisfies the *WA*.

Answer 4.C.11.b. I do not understand this hint at all. I had to compute the entire matrix for person 1 and then could get everything else by symmetry and addition. For consumer 1

$$\begin{aligned}\frac{\partial x_{11}}{\partial p_1} &= -\frac{w}{2p_1^2} - \frac{4}{p_2} \\ \frac{\partial x_{11}}{\partial p_2} &= \frac{4p_1}{p_2^2} \\ \frac{\partial x_{11}}{\partial w} &= \frac{1}{2p_1} \\ \frac{\partial x_{21}}{\partial p_1} &= \frac{8p_1}{p_2^2} \\ \frac{\partial x_{21}}{\partial p_2} &= -\frac{4p_1^2}{p_2^3} \\ \frac{\partial x_{21}}{\partial w} &= 0\end{aligned}$$

Substituting the derivatives to get $S_1(p, w/2)$

$$\begin{aligned}S_1(p, w/2) &= \begin{bmatrix} -\frac{w}{2p_1^2} - \frac{4}{p_2} + \frac{1}{2p_1} \left(\frac{w}{2p_1} - \frac{4p_1}{p_2} \right) & \frac{4p_1}{p_2^2} + \frac{1}{2p_1} \frac{4p_1^2}{p_2^2} \\ \frac{8p_1}{p_2^2} & -\frac{4p_1^2}{p_2^3} \end{bmatrix} \\ &= \begin{bmatrix} -\frac{6}{p_2} & \frac{6p_1}{p_2^2} \\ \frac{8p_1}{p_2^2} & -\frac{4p_1^2}{p_2^3} \end{bmatrix}\end{aligned}$$

Swapping p_1 and p_2 to get $S_2(p, w/2)$

$$S_2(p, w/2) = \begin{bmatrix} -\frac{6}{p_1} & \frac{6p_2}{p_1^2} \\ \frac{8p_2}{p_1^2} & -\frac{4p_2^2}{p_1^3} \end{bmatrix}$$

Summing $S_1(p, w/2)$ and $S_2(p, w/2)$ to get $S(p, w)$

$$S(p, w) = \begin{bmatrix} -\frac{6}{p_2} - \frac{6}{p_1} & \frac{6p_1}{p_2^2} + \frac{6p_2}{p_1^2} \\ \frac{8p_1}{p_2^2} + \frac{8p_2}{p_1^2} & -\frac{4p_1^2}{p_2^3} - \frac{4p_2^2}{p_1^3} \end{bmatrix}$$

Question 4.D.1.

Chapter 5

Question 5.B.1. Draw two production sets: one that violates irreversibility and one that satisfies this property.

Answer 5.B.1.

Question 5.B.2. Suppose that $f(\cdot)$ is the production function associated with a single-output technology, and let Y be the production set of this technology. Show that Y satisfies constant returns to scale if and only if $f(\cdot)$ is homogeneous of degree one.

Answer 5.B.2. Recall that $Y = \{(-z_1, \dots, -z_{L-1}, q) \in \mathbb{R}^L | q - f(z_1, \dots, z_{L-1}) \leq 0 \text{ and } (z_1, \dots, z_{L-1}) \geq 0\}$. Denote $(-z_1, \dots, -z_{L-1}, q)$ by $(-z, q)$.

Suppose first that f is homogeneous of degree 1, and fix a point $(-z, q) \in Y$. Then $q - f(z) \leq 0$ and $z \geq 0$. Then for any $\alpha > 0$, $\alpha q - \alpha f(z) \leq 0$ and $\alpha z \geq 0$. Since f is homogeneous of degree 1, the former condition is equivalent to $\alpha q - f(\alpha z) \leq 0$, i.e. $(-\alpha z, \alpha q) \in Y$ so Y satisfies constant returns to scale.

Now, suppose that Y satisfies constant returns to scale. Then for any point on the transformation frontier there exists a point $(-z, q)$ and any $\alpha > 0$, $\alpha f(z) = \alpha q$. Further, since Y satisfies constant returns to scale, $(-z, q) \in Y$ implies that $(-\alpha z, \alpha q)$ is also on the transformation frontier, so $f(\alpha z) = \alpha q$. Combining these statements, $f(\alpha z) = \alpha f(z)$, i.e. $f(\cdot)$ is homogeneous of degree 1.

Question 5.B.3. Show that for a single-output technology, Y is convex if and only if the production function $f(z)$ is concave.

Answer 5.B.3. Denote $(-z_1, \dots, -z_{L-1}, q)$ by $(-z, q)$. Suppose Y is convex, so that for all $(-z, q), (-z', q') \in Y$ and $\alpha \in [0, 1]$,

$$\alpha(-z, q) + (1 - \alpha)(-z', q') = (-z'', q'') \in Y$$

In particular, this relationship is true when $(-z, q)$ and $(-z', q')$ are on the transformation frontier, so that $f(z) = q$ and $f(z') = q'$. Then by definition of Y

$$\begin{aligned} q'' &\leq f(z'') \\ \alpha q + (1 - \alpha)q' &\leq f(\alpha z + (1 - \alpha)z') \\ \alpha f(z) + (1 - \alpha)f(z') &\leq f(\alpha z + (1 - \alpha)z') \end{aligned}$$

i.e. $f(z)$ is concave.

Now suppose $f(z)$ is concave and that $(-z, q), (-z', q')$ are any two elements of Y , $\alpha \in [0, 1]$ and $(-z'', q'')$ as above. Obviously, $z'' \geq 0$. Since $f(z)$ is concave

$$\alpha f(z) + (1 - \alpha)f(z') \leq f(\alpha z + (1 - \alpha)z')$$

Since $(-z, q), (-z', q') \in Y$, $q \leq f(z)$ and $q' \leq f(z')$. Substituting these inequalities:

$$\begin{aligned} \alpha q + (1 - \alpha)q' &\leq f(z'') \\ q'' &\leq f(z'') \end{aligned}$$

i.e. $(-z'', q'') \in Y$.

Question 5.B.5. Show that if Y is closed and convex, and $-\mathbb{R}_+^L \subset Y$, then free disposal holds.

Answer 5.B.5. Suppose that free disposal does not hold, i.e. there exists $y \in Y$ and $y' \leq y$ with $y' \notin Y$. Since Y is closed and convex, there exists a hyperplane $p \cdot x = w$ separating y' from Y , so that $p \cdot y' > w$ and $p \cdot y \leq w$ for all $y \in Y$. Note in particular that since $-\mathbb{R}_+^L \subset Y$, $0 < y' \leq y$ and thus $p \cdot y' \leq p \cdot y$. Then, by construction, $w < p \cdot y' \leq p \cdot y$, so $w < p \cdot y$, contradicting the separation of Y from y' by $p \cdot x = w$. Therefore, our original hypothesis was wrong, and free disposal does hold.

Question 5.C.1. Prove that, in general, if the production set Y exhibits nondecreasing returns to scale, then either $\pi(p) \leq 0$ or $\pi(p) = +\infty$.

Answer 5.C.1. Suppose $\bar{y}$ solves the firm's profit maximization problem (i.e. $p \cdot \bar{y} = \pi(p)$)

$$\max_y p \cdot y \text{ subject to } F(y) \leq 0$$

Suppose first that $\pi(p) > 0$. Then for any $\alpha \geq 1$

$$p \cdot \alpha \bar{y} > \pi(p) \text{ and } F(\alpha \bar{y}) \leq \alpha F(\bar{y})$$

(Recall that $F(\alpha \bar{y}) = \alpha F(\bar{y})$ is the constant returns to scale condition by Exercise 5.B.2.) These conditions imply that in fact $\bar{y}$ does not solve the PMP; rather $\alpha \bar{y}$ is an improvement. Since α is unbounded above, $\pi(p)$ is unbounded above, i.e. $\pi(p) = +\infty$.

Alternatively, suppose that $\pi(p) \leq 0$. Then for any $\alpha \geq 1$

$$p \cdot \alpha \bar{y} \leq \pi(p) \text{ and } F(\alpha \bar{y}) \leq \alpha F(\bar{y})$$

In this case, the above statement confirms that $\bar{y}$ does solve the PMP and $\pi(p) \leq 0$.

Question 5.C.2. Prove that $\pi(\cdot)$ is a convex function. [Property (ii) of Proposition 5.C.1]. [Hint: Suppose that $y \in y(\alpha p + (1 - \alpha)p')$. Then

$$\pi(\alpha p + (1 - \alpha)p') = \alpha p \cdot y + (1 - \alpha)p' \cdot y \leq \alpha \pi(p) + (1 - \alpha)\pi(p')$$

Answer 5.C.2. This solution is just a fleshing-out of the hint.

$$\begin{aligned} \pi(\alpha p + (1 - \alpha)p') &= (\alpha p + (1 - \alpha)p') \cdot y \\ &= \alpha p \cdot y + (1 - \alpha)p' \cdot y \end{aligned}$$

Since $y \in y(p)$ implies y is a profit maximizing choice given p , it follows that for any arbitrary y , $p \cdot y \leq \pi(p)$. Therefore

$$\pi(\alpha p + (1 - \alpha)p') \leq \alpha \pi(p) + (1 - \alpha)\pi(p')$$

Question 5.C.3. Establish properties (viii) and (ix) of Proposition 5.C.2. [Hint: Property (viii) is easy; (ix) is more difficult. Try the one-input case first.]

Answer 5.C.3. Property (viii): If $f(z)$ is homogeneous of degree 1, then $z(w, q)$ and $c(w, q)$ are homogeneous of degree 1 in q . First, we show that $z(w, q)$ is homogeneous of degree 1 in q . Suppose that $(\bar{z}_1, \dots, \bar{z}_{L-1})$ solves

$$\min_{z \geq 0} w \cdot z \text{ subject to } f(z) \geq q$$

In other words, $(\bar{z}_1, \dots, \bar{z}_{L-1})$ solves

$$\begin{aligned} \bar{z}_l \left(w_l - \lambda \frac{\partial f}{\partial z} \Big|_{(\bar{z}_1, \dots, \bar{z}_{L-1})} \right) &= 0 \\ f(\bar{z}_1, \dots, \bar{z}_{L-1}) &\geq q \end{aligned}$$

Since $f(z)$ is homogeneous of degree 1, the partials of $f(z)$ are homogeneous of degree 0. Therefore we can rewrite the above system as

$$\alpha \bar{z}_l \left(w_l - \lambda \frac{\partial f}{\partial z} \Big|_{(\alpha \bar{z}_1, \dots, \alpha \bar{z}_{L-1})} \right) = 0$$

$$f(\alpha \bar{z}_1, \dots, \alpha \bar{z}_{L-1}) \geq \alpha q$$

Since these equations are exactly the conditions that say $(\alpha \bar{z}_1, \dots, \alpha \bar{z}_{L-1})$ solves

$$\min_{z \geq 0} w \cdot z \text{ subject to } f(z) \geq \alpha q$$

we can conclude that $z(w, \alpha q) = \alpha z(w, q)$, i.e. $z(w, q)$ is homogeneous of degree 1 in q . Now, it follows immediately that $c(w, q)$ is homogeneous of degree 1 in q since $c(w, \alpha q) = w \cdot z(w, \alpha q) = \alpha w \cdot z(w, q) = \alpha c(w, q)$.

Property (ix): If $f(z)$ is concave, then $c(\cdot)$ is a convex function of q (in particular, marginal costs are nondecreasing in q). Assuming that $\{0\} \in Y \subseteq \mathbb{R}^L$, then the concavity of f implies that Y satisfies non-increasing returns to scale. Fix a price level w and let q and q' denote any two output levels feasible at w . Then for any $\alpha \in [0, 1]$

$$f(\alpha z(w, q)) \geq \alpha q = f(z(w, \alpha q))$$

$$f((1 - \alpha)z(w, q')) \geq (1 - \alpha)q' = f(z(w, (1 - \alpha)q'))$$

Summing

$$f(\alpha z(w, q)) + f((1 - \alpha)z(w, q')) \geq \alpha q + (1 - \alpha)q' = f(z(w, \alpha q)) + f(z(w, (1 - \alpha)q'))$$

$$f(\alpha z(w, q)) + f((1 - \alpha)z(w, q')) \geq f(z(w, \alpha q + (1 - \alpha)q'))$$

Therefore, at any fixed price level

$$w \cdot (\alpha z(w, q) + (1 - \alpha)z(w, q')) \geq w \cdot (z(w, \alpha q + (1 - \alpha)q'))$$

$$\alpha c(w, q) + (1 - \alpha)c(w, q') \geq c(w, \alpha q + (1 - \alpha)q')$$

Question 5.C.4. Establish properties (i) to (vii) of Proposition 5.C.2 for the case in which there are multiple outputs.

Answer 5.C.4. When we write $f(z) \geq q$, now consider q a vector in $\mathbb{R}_+^m$ with all notation as usual.

- (1) $c(\cdot)$ is homogeneous of degree 1 in w and nondecreasing in q

Since $z(w, q)$ is homogeneous of degree 0, for any $\alpha > 0$ $\alpha c(w, q) = \alpha w \cdot z(w, q) = \alpha w \cdot z(\alpha w, q) = c(\alpha w, q)$, so $c(\cdot)$ is homogeneous of degree 1 in w .

- (2) $c(\cdot)$ is a concave function of w . Suppose $w'' = \alpha w + (1 - \alpha)w'$, $\alpha \in [0, 1]$, and suppose that z is some cost-minimizing input vector. Then

$$\begin{aligned} c(w'', q) &= w'' \cdot z \\ &= \alpha w \cdot z + (1 - \alpha)w' \cdot z \\ &\geq \alpha c(w, q) + (1 - \alpha)c(w', q) \end{aligned}$$

since $w \cdot z \geq c(w, q)$ by the definition of the CMP.

- (3) If the sets $\{z \mid f(z) \geq q\}$ are convex for every q , then $Y = \{(-z, q) \mid w \cdot z \geq c(w, q) \text{ for all } w \gg 0\}$.
- (4) $z(\cdot)$ is homogeneous of degree 0 in w

For any $\alpha > 0$

$$\begin{aligned} z(\alpha w, q) &= \underset{z}{\operatorname{argmin}} \alpha w \cdot z \text{ subject to } f(z) \geq q \\ z(w, q) &= \underset{z}{\operatorname{argmin}} w \cdot z \text{ subject to } f(z) \geq q \end{aligned}$$

Since the constraint sets are the same in these two problems and the objective functions are strictly increasing transformations of each other, they have the same argument at their respective minima, so $z(\alpha w, q) = z(w, q)$.

(5)

(6)

(7)

Question 5.C.5. Argue that for property (iii) of Proposition 5.C.2 to hold, it suffices that $f(\cdot)$ be quasiconcave. Show that the quasiconcavity of $f(\cdot)$ is compatible with increasing returns.

Answer 5.C.5. Property (iii) of Proposition 5.C.2 requires that the set $\{z \geq 0 | f(z) \geq q\}$ be convex for all q . These sets are exactly the upper contour sets of f , which are all convex exactly when f is quasiconcave.

Suppose that $f(z) = \ln z$, a quasiconcave function. Let $\alpha > 1$, and $\ln z \geq q$.

Question 5.C.8. Alpha Incorporated (AI) produces a single output q from two inputs z_1 and z_2 . You are assigned to determine AI's technology. You are given 100 monthly observations. Two of these monthly observations are shown in the following table:

	Input prices		Input levels		Output price	Output level
Month	w_1	w_2	z_1	z_2	p	q
3	3	1	40	50	4	60
95	2	2	55	40	4	60

In light of these two monthly observations, what problem will you encounter in trying to accomplish your task?

Answer 5.C.8. The firm's output is the same in both periods. However, its cost in month 95 is $2 * 55 + 2 * 40 = 190$, but the cost of the month 3 inputs at month 95 prices is $2 * 40 + 2 * 50 = 180$, so the firm did not solve its cost minimization in month 95.

Question 5.C.9. Derive the profit function $\pi(p)$ and supply function (or correspondence) $y(p)$ for the single-output technologies whose production functions $f(z)$ are given by

Question 5.C.9.a. $f(z) = \sqrt{z_1 + z_2}$

Answer 5.C.9.a.

$$\max_{z_1 \geq 0, z_2 \geq 0} p\sqrt{z_1 + z_2} - w_1 z_1 - w_2 z_2$$

First order conditions

$$\begin{aligned} \frac{p}{2\sqrt{z_1 + z_2}} - w_1 &\leq 0, = 0 \text{ if } z_1 > 0 \\ \frac{p}{2\sqrt{z_1 + z_2}} - w_2 &\leq 0, = 0 \text{ if } z_2 > 0 \end{aligned}$$

Note that production is symmetric across the inputs, so if $w_1 = w_2$, the is essentially producing one output with one input. If $w_1 > w_2$, the non-negativity constraint on z_1 binds. If $w_2 > w_1$, the non-negativity constraint on z_2 binds. Combining these conclusions

$$y(p, w_1, w_2) = (-z_1, -z_2, q) = \left(-\frac{\alpha p^2}{4w_1^2}, -\frac{(1-\alpha)p^2}{4w_2^2}, \frac{p}{2\min\{w_1, w_2\}} \right)$$

where $\alpha = 0$ if $w_1 > w_2$, $\alpha = 1$ if $w_1 < w_2$ and $\alpha \in [0, 1]$ if $w_1 = w_2$. Then

$$\pi(p, w_1, w_2) = (p, w_1, w_2) \cdot y(p, w_1, w_2) = \frac{p^2}{4\min\{w_1, w_2\}}$$

Question 5.C.9.b. $f(z) = \sqrt{\min\{z_1, z_2\}}$

Answer 5.C.9.b. Graphing the iso-output curves, we see that the CMP is solved when $z_1 = z_2 = z$. Since the PMP is the dual of the CMP, the PMP is

$$\max_{z \geq 0} p\sqrt{z} - z(w_1 + w_2)$$

First order condition

$$\frac{p}{2\sqrt{z}} - (w_1 + w_2) \leq 0, \quad = 0 \text{ if } z > 0$$

At $z = 0$, profit is 0, so any interior is better if profit is positive. The interior solution is

$$z = \left(\frac{p}{2(w_1 + w_2)} \right)^2$$

Supply correspondence

$$y(p, w_1, w_2) = (-z_1, -z_2, q) = \left(-\frac{p^2}{4(w_1 + w_2)^2}, -\frac{p^2}{4(w_1 + w_2)^2}, \frac{p}{2(w_1 + w_2)} \right)$$

Profit function

$$\pi(p, w_1, w_2) = (p, w_1, w_2) \cdot y(p, w_1, w_2) = \frac{p^2}{4(w_1 + w_2)}$$

Question 5.C.9.c. $f(z) = (z_1^\rho + z_2^\rho)^{1/\rho}$ for $\rho \leq 1$

Answer 5.C.9.c.

Question 5.D.1.

Question 5.E.1.

Question 5.F.1.

Question 5.G.1.

Question 5.AA.1.

Chapter 6

Chapter 7

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Chapter 9

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Chapter 11

Chapter 12

Question 12.B.1.

Question 12.C.1.

Question 12.D.1.

Question 12.E.1.

Question 12.F.1.

Question 12.G.1.

Question 12.AA.1.

Question 12.BB.1.

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Chapter 14

Chapter 15

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Chapter 19

Chapter 20

Chapter 21**Chapter 22****Chapter 23**

Note: I have tried to give credit to the people who have helped me solve these problems, at times just by showing me a solution. However, if you notice an error in giving credit, please let me know so I can fix it. If you notice any errors in the solutions themselves, please let me know so that I can fix it. If you have a solution to a problem that is not solved herein, or a different solution to one that is, I would be most grateful if you would send it my way, so that I can add it. All errors, incompleteness, etc. are mine.